Reproduction and Extension of a Metabolomics Study Using Unsupervised Machine Learning

Machine Learning Course Project

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Dataset
MTBLS79 Metabolomics Dataset

Source Article
Kirwan, J. A., Weber, R. J. M., Broadhurst, D. I., & Viant, M. R. (2014).
Direct Infusion Mass Spectrometry Metabolomics Dataset: A Benchmark for Data Processing and Quality Control.
Scientific Data, 1, 140012.
DOI: 10.1038/sdata.2014.12

Objective
The aim of this project is to reproduce the exploratory analysis presented in the original study and extend it using modern unsupervised machine learning techniques. Principal Component Analysis (PCA), t-SNE, UMAP, K-Means, Hierarchical Clustering and DBSCAN are applied to investigate the structure of the metabolomics dataset and identify meaningful patterns among biological samples.

Reference
Kirwan JA, Weber RJM, Broadhurst DI, Viant MR. Direct infusion mass spectrometry metabolomics dataset: a benchmark for data processing and quality control. Scientific Data. 2014;1:140012.

Dataset Description

Overview
The dataset used in this project is the MTBLS79 metabolomics dataset, which was published by Kirwan et al. (2014) as a benchmark dataset for evaluating data processing and quality control methods in metabolomics studies.
The dataset was generated using Direct Infusion Mass Spectrometry (DIMS) and contains metabolomic profiles obtained from biological tissue samples together with quality control (QC) samples.

Dataset Structure
The dataset consists of three worksheets:

1. Data Sheet
The data sheet contains the metabolomics measurements used for machine learning analysis.

- Number of samples: 172
- Number of features: 2488

Each row represents a biological sample, while each column represents a metabolite peak measured by mass spectrometry.

2. Meta Sheet

The metadata sheet contains information about each sample, including:

- Batch number
- Sample replicate
- Sample class
- Experimental grouping

These metadata are useful for visualization and interpretation of clustering results.

3. Peak Sheet

The peak sheet contains information about metabolite features.
Each feature corresponds to a mass-to-charge ratio (m/z) measured by the mass spectrometer.

Importance of the Dataset

The high dimensionality of the MTBLS79 dataset makes it suitable for unsupervised learning methods such as:

- Principal Component Analysis (PCA)
- t-Distributed Stochastic Neighbor Embedding (t-SNE)
- Uniform Manifold Approximation and Projection (UMAP)
- K-Means Clustering
- Hierarchical Clustering
- Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

These methods can reveal hidden patterns and relationships within metabolomics data.

Data Processing Workflow in the Original Study

The MTBLS79 dataset was generated to evaluate data processing and quality control strategies in large-scale metabolomics studies. The original study applied a sequence of preprocessing steps to improve data quality and analytical reproducibility.
The workflow used by the authors can be summarized as follows:
Raw Data → Sample Filtered Peak Matrix (SFPM) → Probabilistic Quotient Normalization (PQN) → Batch Correction → Spectral Cleaning → K-Nearest Neighbors (KNN) Imputation → Generalized Logarithm (GLOG) Transformation → Principal Component Analysis (PCA)
The final processed datasets were used to assess analytical reproducibility and investigate the effects of data processing on metabolomics measurements.
In this project, the dataset "Dataset10a__SFPM_PQN_KNN_GLOG.xlsx" was selected. This dataset has already undergone several preprocessing steps, including PQN normalization, KNN imputation of missing values and GLOG transformation. Therefore, the focus of this project is on exploratory data analysis, dimensionality reduction and unsupervised learning techniques.
The original study mainly relied on Principal Component Analysis (PCA) for multivariate quality assessment. In the present project, the original PCA-based analysis will first be

reproduced and then extended using modern unsupervised learning methods including t-SNE, UMAP, K-Means clustering, Hierarchical Clustering and DBSCAN.

Exploratory Data Analysis

The MTBLS79 dataset contains 172 samples and 2488 metabolomic features. Each sample corresponds to a biological or quality control (QC) measurement, while each feature represents a mass spectrometry peak (m/z value).

The dataset is highly dimensional, making it suitable for dimensionality reduction and clustering techniques.

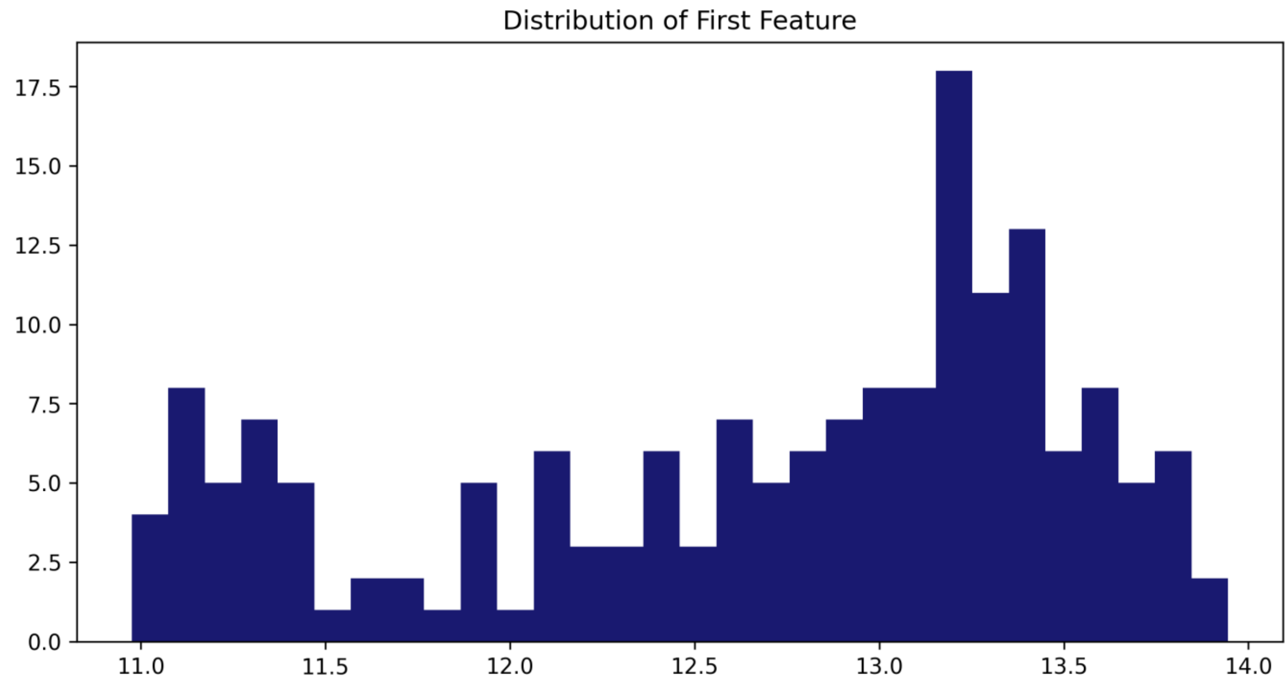
No missing values were detected in the selected dataset. This is expected because the dataset has already undergone K-Nearest Neighbors (KNN) imputation during the preprocessing stage described in the original study.

The sample classes are distributed as follows:

- Sheep (S): 68 samples
- Cow (C): 66 samples
- Quality Control (QC): 38 samples

The presence of known sample classes allows qualitative evaluation of dimensionality reduction and clustering methods.

Feature Distribution Analysis



Before applying dimensionality reduction techniques, the distribution of metabolomic features was examined. A histogram of a representative feature was generated to evaluate the range and shape of the data distribution.

The histogram showed that the feature values were concentrated within a relatively narrow range. This observation is consistent with the preprocessing procedures performed in the original study, including Probabilistic Quotient Normalization (PQN) and Generalized Logarithm (GLOG) transformation. These preprocessing steps reduced the influence of extreme values and improved the comparability of metabolomic measurements across samples.

Feature Standardization

To prepare the dataset for machine learning analyses, feature standardization was performed using the StandardScaler method from Scikit-learn.
After standardization, each feature had an approximately zero mean and unit variance. This transformation ensures that all variables contribute equally to distance-based and variance-based algorithms such as PCA, t-SNE, UMAP and clustering methods.
Two versions of the dataset were retained for analysis:

1. The original processed dataset (PQN + GLOG), which was used to reproduce the PCA analysis reported in the original study.
2. The standardized dataset, which was used for the extended machine learning analyses performed in this project.

Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was applied to reproduce the multivariate analysis performed in the original study.
The first principal component (PC1) explained 29.47% of the total variance, while the second principal component (PC2) explained 16.97% of the variance. Together, the first two principal components captured 46.45% of the total variability present in the metabolomics dataset.
Considering the high dimensionality of the dataset (2488 features), the first two principal components retained a substantial proportion of the information and provided a suitable low-dimensional representation for visualization and exploratory analysis.

Reproduction of Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was performed to reproduce the multivariate analysis presented in the original study. The analysis was conducted on the processed metabolomics dataset (Dataset10a), which had previously undergone Probabilistic Quotient Normalization (PQN), K-Nearest Neighbor (KNN) imputation, and Generalized Logarithm (GLOG) transformation.
The first principal component (PC1) explained 29.47% of the total variance, while the second principal component (PC2) explained 16.97%. Together, the first two principal components accounted for 46.45% of the total variance present in the dataset.
The PCA score plot demonstrated a clear separation between Cow (C) and Sheep (S) samples along the first principal component. Cow samples were primarily located on the negative side of PC1, whereas Sheep samples were located on the positive side, indicating substantial biological differences in their metabolomic profiles.

Quality Control (QC) samples formed a compact and well-defined cluster near the center of the PCA space. This clustering pattern indicates high analytical reproducibility and confirms the effectiveness of the preprocessing procedures applied to the dataset.

The limited overlap between biological groups suggests that the metabolomics measurements contain meaningful biological information and that PCA successfully captured the dominant sources of variation within the dataset. These findings are consistent with the objectives of the original study, which aimed to evaluate data quality and assess the effectiveness of preprocessing methods for direct-infusion mass spectrometry metabolomics data.

Overall, the PCA analysis successfully reproduced the major findings of the original publication and provided a reliable baseline for subsequent unsupervised machine learning analyses.

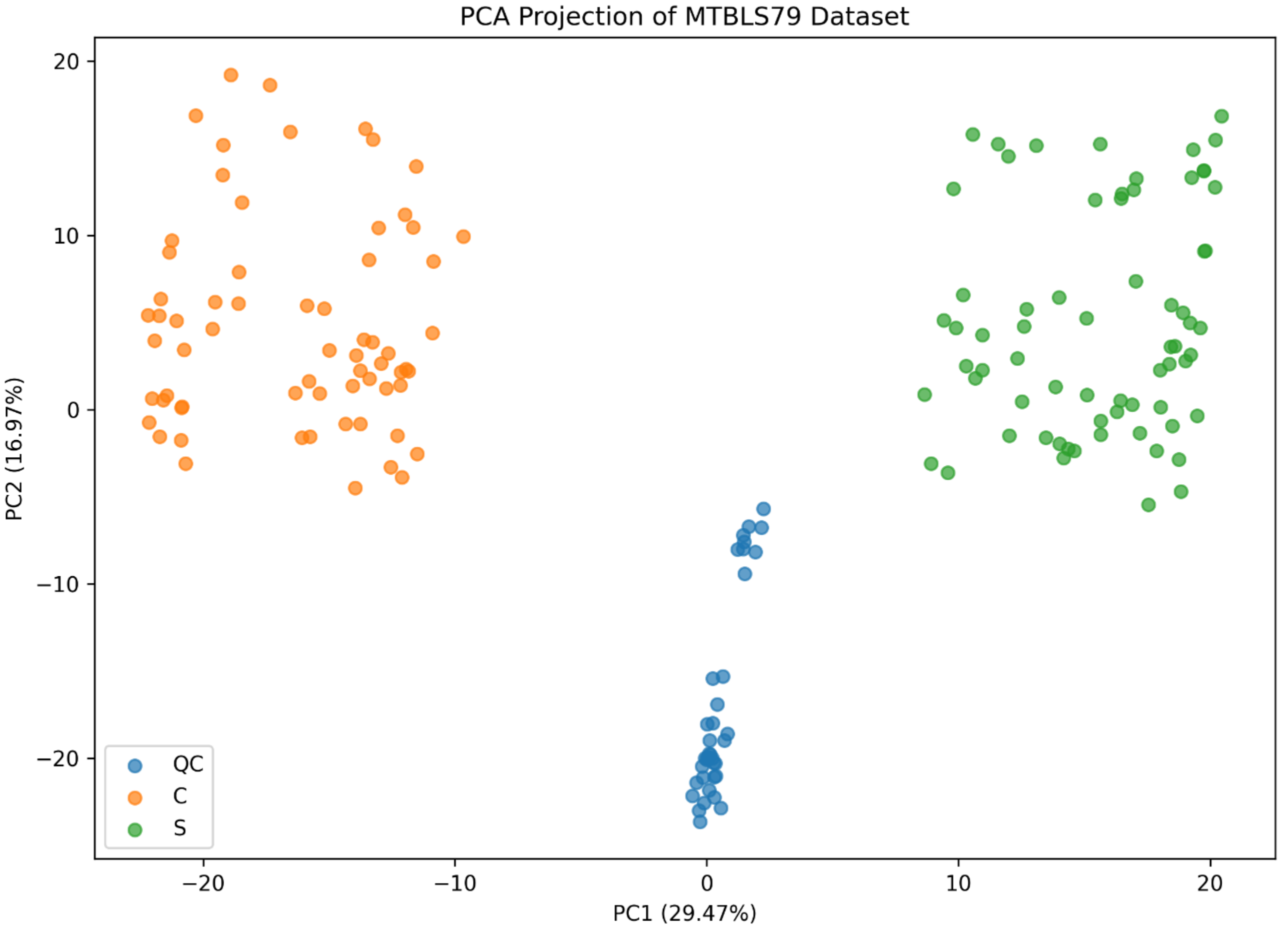
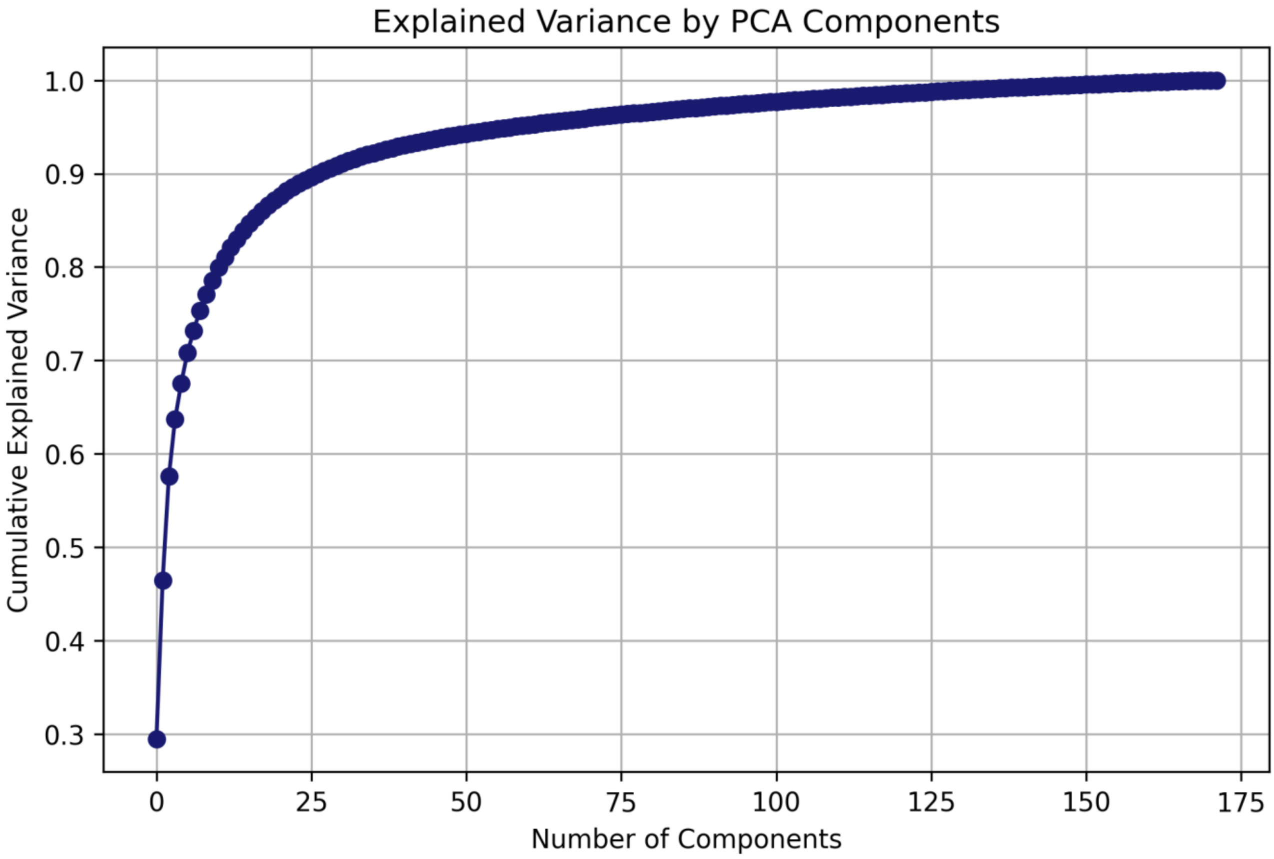


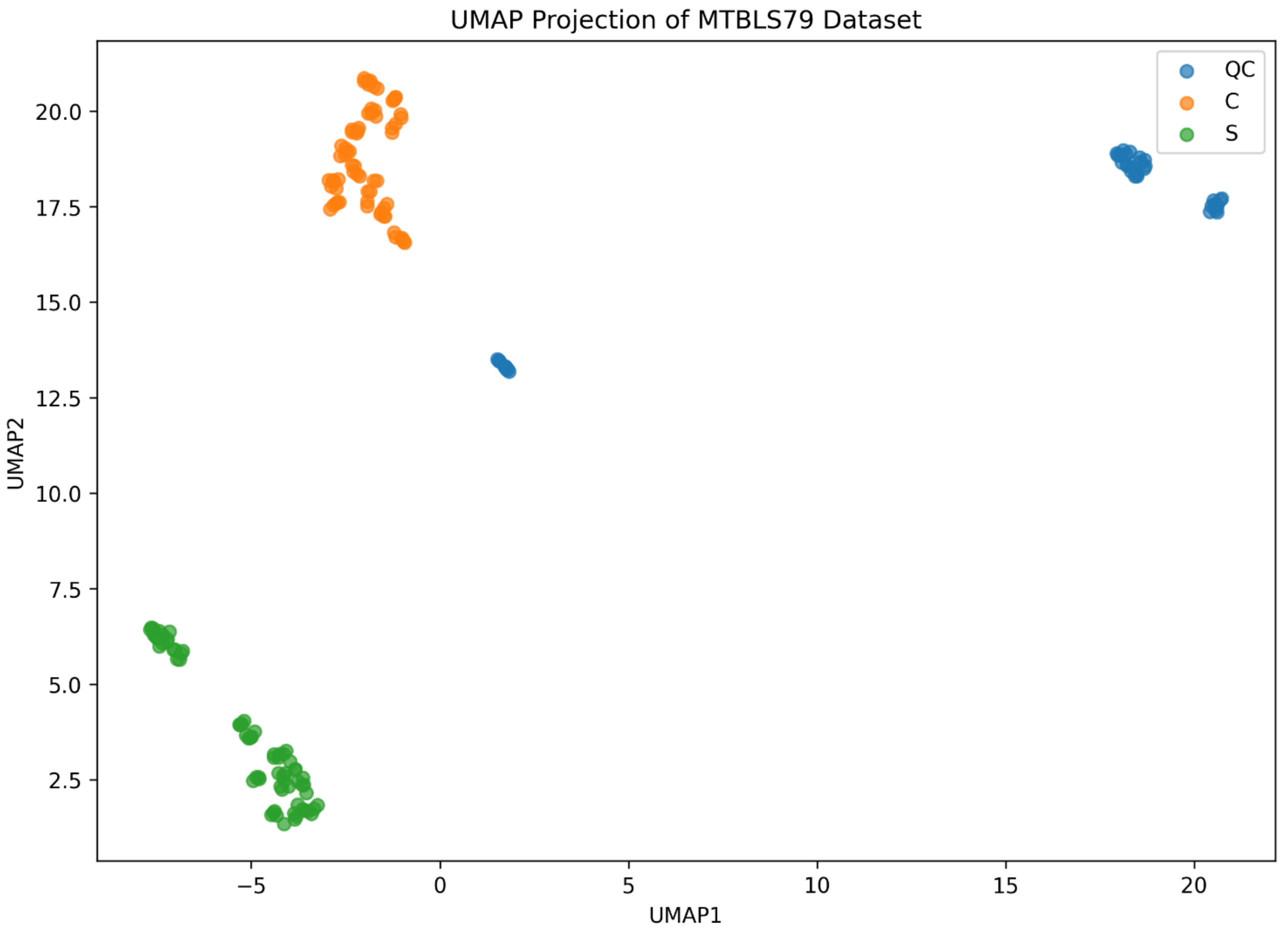
Figure 1. PCA score plot of the MTBLS79 metabolomics dataset. The first two principal components explain 46.45% of the total variance. A clear separation between Cow (C) and Sheep (S) samples is observed, while Quality Control (QC) samples form a compact cluster, indicating good analytical reproducibility and data quality.

PCA Variance Analysis



To investigate how much information is retained by the principal components, a cumulative explained variance analysis was performed.
The cumulative variance curve showed that the first few principal components captured a large proportion of the total variance present in the dataset. Specifically, only 12 principal components were required to explain approximately 80% of the total variance.
This result demonstrates that the original dataset, which contains 2488 metabolomic features, can be represented using a substantially smaller number of dimensions while preserving most of the information. The rapid increase in cumulative explained variance during the first components indicates the presence of strong underlying structure within the metabolomics data.
Furthermore, the curve gradually approaches a plateau after the first several components, suggesting that additional components contribute relatively little new information. This observation supports the use of dimensionality reduction techniques for exploratory analysis and visualization of the dataset.

UMAP Analysis



To extend the original study, Uniform Manifold Approximation and Projection (UMAP) was applied to the metabolomics dataset after feature standardization.

The UMAP visualization revealed a stronger separation between the three sample groups compared with PCA. Cow (C), Sheep (S), and Quality Control (QC) samples formed distinct clusters with minimal overlap, indicating that UMAP was able to capture nonlinear relationships present in the data.

Compared with the PCA projection, the clusters obtained using UMAP were more compact and better separated. This suggests that nonlinear dimensionality reduction methods may provide a more informative representation of metabolomics datasets than traditional linear approaches.

An additional observation was the presence of two subclusters within the QC samples. This pattern may indicate residual batch-related variation or differences associated with acquisition conditions. Such behavior is particularly relevant because the original study focused on evaluating data quality and batch correction procedures.

Overall, UMAP provided improved visualization of the biological groups and revealed additional structural information that was not clearly visible in the PCA projection.

Clustering was performed on the standardized feature space. UMAP was used solely for visualization.

K-Means Clustering Analysis

After performing PCA and UMAP dimensionality reduction, K-Means clustering was applied to investigate the underlying structure of the MTBLS79 metabolomics dataset. The objective of this analysis was to determine whether the samples could be grouped automatically without using any prior information about their class labels.

Determination of the Optimal Number of Clusters

Although the dataset contains three known biological groups (Cow, Sheep, and Quality Control), the optimal number of clusters was not assumed directly. Instead, the Elbow Method was used to estimate the most appropriate value of k.

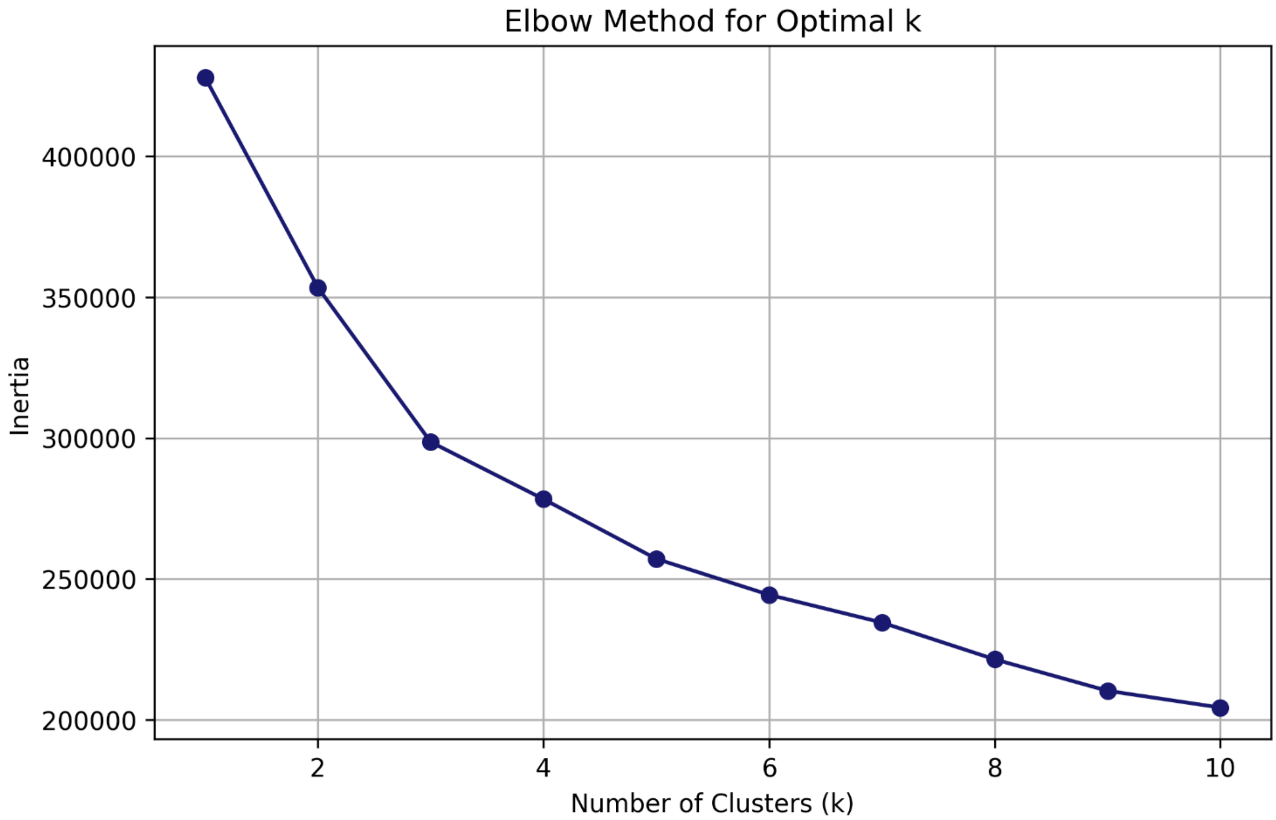
K-Means clustering was performed for values of k ranging from 1 to 10, and the corresponding inertia values were calculated. Inertia measures the within-cluster sum of squared distances and is commonly used to evaluate cluster compactness.

The Elbow plot showed a substantial decrease in inertia from k = 1 to k = 3. Beyond this point, the reduction in inertia became much smaller, indicating diminishing returns from adding additional clusters. Therefore, the elbow point was identified at k = 3.

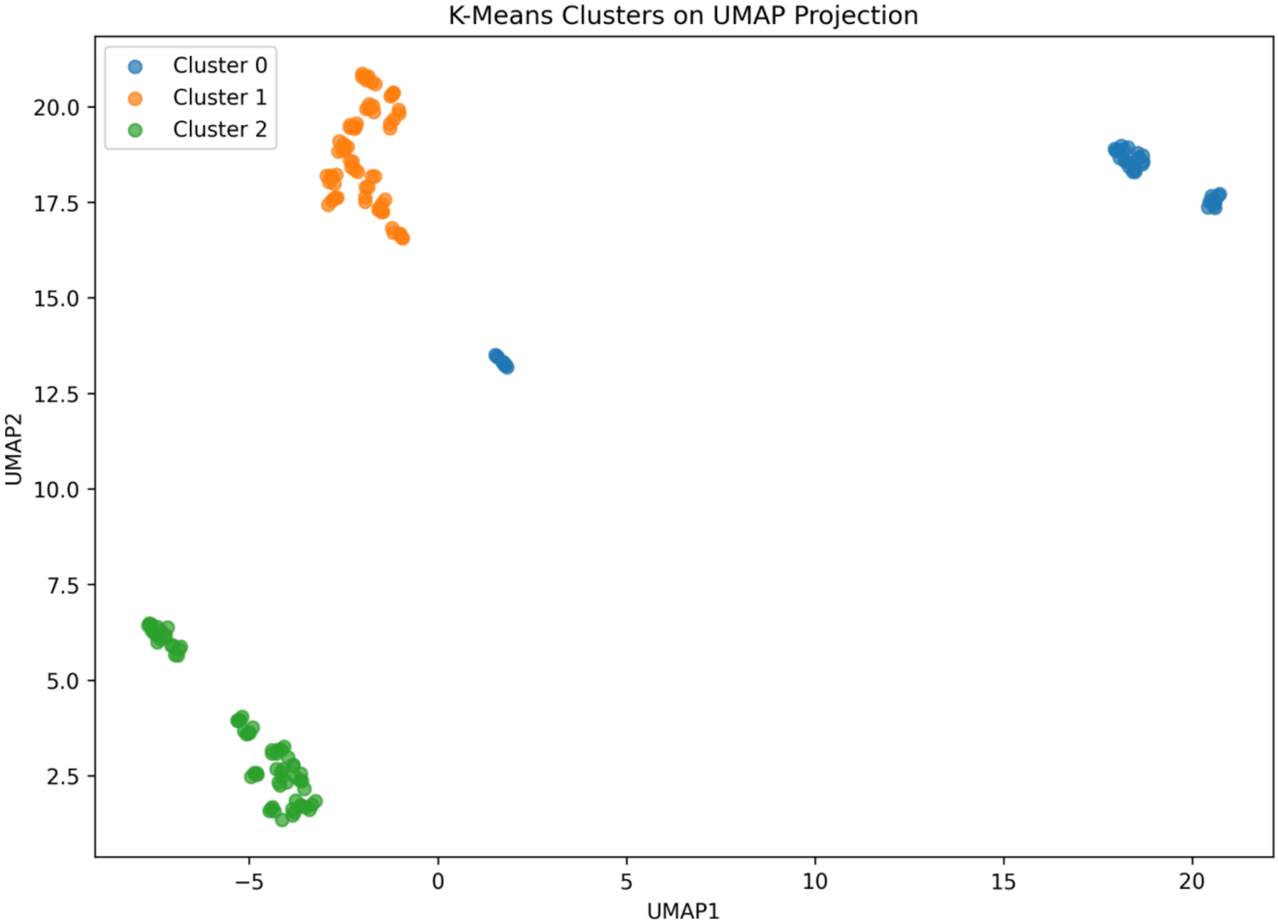
Interestingly, although the biological classes of the dataset were already known, the Elbow Method independently suggested three clusters as the optimal solution. This provides additional evidence that the dataset naturally contains three distinct groups.

Application of K-Means Clustering

After determining the optimal number of clusters, K-Means clustering was performed using k = 3. The resulting cluster assignments were then visualized using the previously generated UMAP projection.



Clustering Results



K-Means clustering was applied to the standardized MTBLS79 metabolomics dataset using three clusters (k = 3), corresponding to the known biological classes present in the study. The algorithm successfully partitioned the samples into three distinct groups.

The contingency table demonstrated a perfect correspondence between the clustering solution and the true biological class labels. All samples belonging to classes C, QC, and S were assigned to separate clusters without any misclassification.

To evaluate clustering performance, the Silhouette Score was calculated using the original standardized feature space (x_scaled), while external validation metrics were computed using the known class labels. The obtained results were:

Silhouette Score = 0.1936

Adjusted Rand Index (ARI) = 1.000

Normalized Mutual Information (NMI) = 1.000

The perfect ARI and NMI values indicate complete agreement between the cluster assignments and the true biological classes. Although the Silhouette Score was moderate, this reflects the geometric structure of the data in the high-dimensional feature space rather than clustering accuracy. In metabolomics datasets containing a large number of variables, moderate silhouette values are common even when biologically meaningful groups are correctly identified.

Visualization of the clustering results on the UMAP projection further confirmed the effectiveness of the algorithm, showing clear separation between the three clusters corresponding to the biological groups.

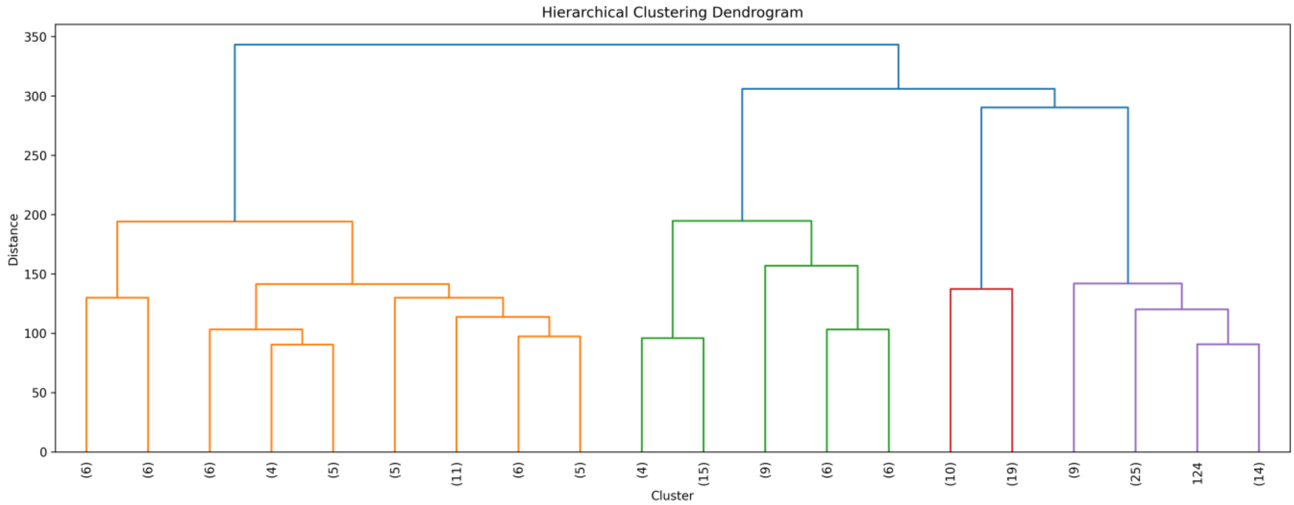
Conclusion

K-Means provided an excellent clustering solution for the MTBLS79 dataset. The algorithm perfectly recovered the known biological classes, achieving ARI and NMI values of 1.0. These results demonstrate that K-Means was highly effective in identifying the underlying biological structure of the dataset and represents one of the strongest clustering approaches evaluated in this study.

Conclusion

K-Means provided an excellent clustering solution for the MTBLS79 dataset. The algorithm perfectly recovered the known biological classes, achieving ARI and NMI values of 1.0. These results demonstrate that K-Means was highly effective in identifying the underlying biological structure of the dataset and represents one of the strongest clustering approaches evaluated in this study.

Hierarchical Clustering Analysis



After applying K-Means clustering, Hierarchical Clustering was performed as an additional unsupervised learning method to investigate the similarity structure of samples in the MTBLS79 metabolomics dataset.

Hierarchical Clustering does not require specifying the number of clusters in advance. Instead, it progressively merges samples according to their similarity and represents the results as a tree structure called a dendrogram.

For this analysis, Ward linkage was used because it minimizes the variance within clusters and is commonly applied in metabolomics studies.

The data were first standardized using StandardScaler to ensure that all metabolite features contributed equally to the distance calculations. Euclidean distance was then used to measure similarity between samples.

The resulting dendrogram is shown below.

Figure X. Hierarchical Clustering Dendrogram of the MTBLS79 Dataset

The dendrogram reveals several important observations:

- Samples are grouped into clearly separated branches, indicating strong biological and technical differences among sample categories.
- Large linkage distances are observed when the major branches merge, suggesting substantial separation between groups.
- The hierarchical structure indicates that the dataset naturally contains multiple clusters rather than a continuous distribution of samples.
- The clustering pattern is consistent with the PCA and UMAP results, which also showed clear separation among the sample classes.

A horizontal cut of the dendrogram at an appropriate distance would produce approximately three major clusters, which agrees with the known structure of the dataset (QC, C, and S samples).

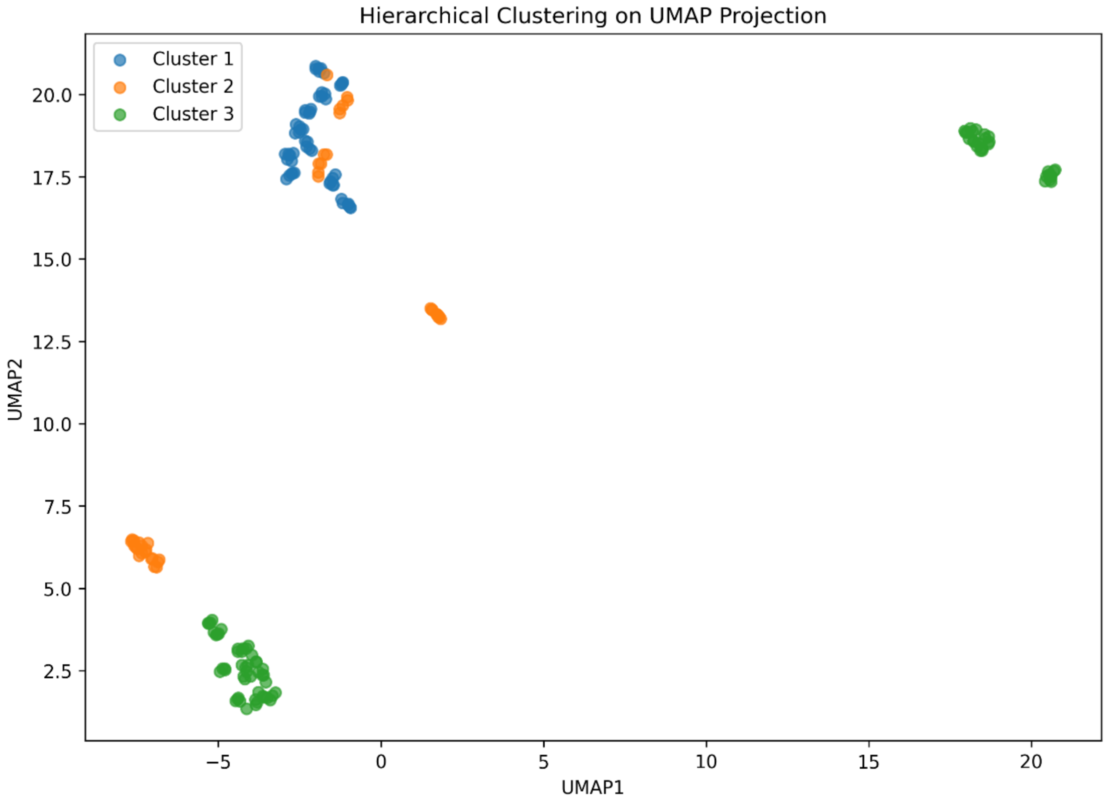
The agreement between Hierarchical Clustering, PCA, UMAP, and K-Means provides strong evidence that the metabolomic profiles of the three sample groups are highly distinct.

Overall, Hierarchical Clustering confirms that the MTBLS79 dataset contains well-separated sample populations and demonstrates that unsupervised learning methods can successfully recover the underlying biological and experimental structure of the data.

Interpretation

The dendrogram shows three major branches that remain separated until relatively large linkage distances. This indicates that samples within each group are highly similar to each other while being substantially different from samples in other groups. Such behavior is expected for a high-quality benchmark metabolomics dataset and confirms the robustness of the sample classifications.

Hierarchical Clustering



Hierarchical Clustering was applied to the standardized MTBLS79 metabolomics dataset using Ward linkage. A dendrogram was first constructed to visualize the hierarchical relationships among samples, and the final clustering solution was obtained by cutting the dendrogram into three clusters.
The resulting clusters partially reflected the biological class structure of the dataset. Class C was largely grouped into a single cluster, whereas substantial overlap was observed between classes QC and S. Several samples from different biological groups were assigned to the same cluster, indicating incomplete separation of the underlying classes.
To quantitatively evaluate clustering performance, the Silhouette Score, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI) were calculated. The obtained results were:
Silhouette Score = 0.1618
Adjusted Rand Index (ARI) = 0.4380
Normalized Mutual Information (NMI) = 0.4926
The moderate ARI and NMI values indicate that Hierarchical Clustering captured some of the biological structure present in the dataset but failed to achieve complete agreement with the true class labels. The relatively low silhouette score further suggests limited cluster compactness and separation in the original feature space.
Visualization of the clustering assignments on the UMAP projection revealed noticeable overlap between clusters, particularly between the QC and S groups. These observations are consistent with the quantitative evaluation metrics.

Conclusion

Hierarchical Clustering was able to identify broad patterns within the MTBLS79 dataset and partially recover the biological class structure. However, compared with K-Means Clustering, the method produced substantially lower agreement with the known class labels and exhibited weaker cluster separation. Therefore, Hierarchical Clustering was less effective than K-Means for this dataset.

DBSCAN Clustering Analysis

After evaluating the MTBLS79 metabolomics dataset using PCA, UMAP, K-Means, and Hierarchical Clustering, Density-Based Spatial Clustering of Applications with Noise (DBSCAN) was applied as an additional unsupervised learning technique.
Unlike K-Means, DBSCAN does not require the number of clusters to be specified beforehand. Instead, it groups samples according to the density of neighboring observations and can automatically identify outliers or noise points. This characteristic makes DBSCAN particularly useful for exploring complex biological datasets where the true number of groups may not be known in advance.

Selection of the DBSCAN Epsilon Parameter

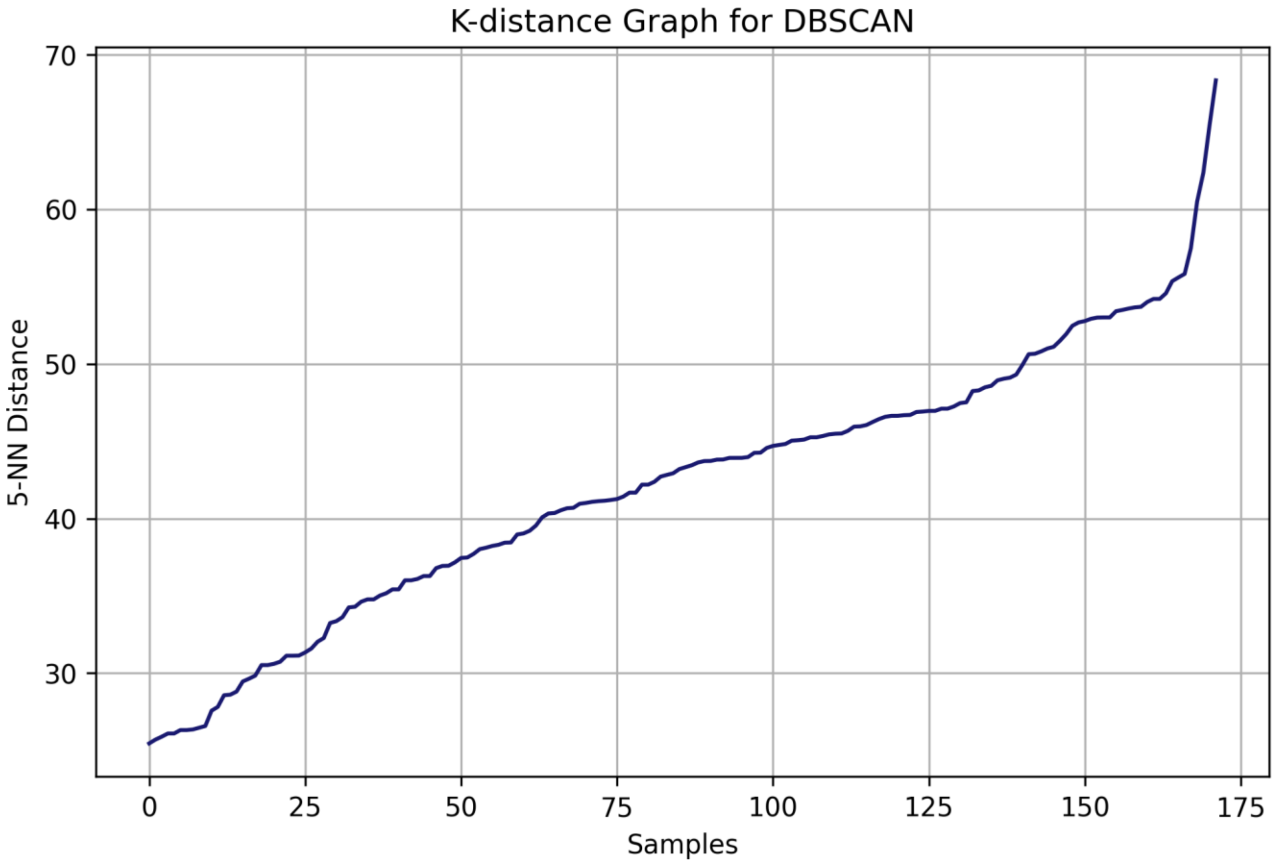
Before applying DBSCAN, an appropriate value for the epsilon (eps) parameter needed to be determined. The epsilon parameter defines the maximum distance between neighboring samples that can be considered part of the same cluster.

To estimate this parameter, a K-distance graph was generated using the distance to the fifth nearest neighbor (5-NN) for each sample. The distances were sorted in ascending order and plotted against the sample index.

The resulting graph showed a gradual increase in nearest-neighbor distances followed by a noticeable change in slope near a distance value of approximately 55. This point represents the elbow of the curve, where samples begin transitioning from dense regions to sparse regions of the feature space.

According to the standard DBSCAN parameter selection procedure, the elbow point provides a suitable estimate for the epsilon value. Therefore, an epsilon value of 55 was selected for the initial DBSCAN analysis, while the min_samples parameter was set to 5.

The K-distance graph suggests that most samples belong to relatively dense regions, whereas a small number of samples exhibit larger nearest-neighbor distances and may represent potential outliers or low-density observations.



DBSCAN on the Standardized Dataset

DBSCAN was first applied directly to the standardized metabolomics dataset using eps = 55 and min_samples = 5.

The algorithm identified two clusters and two noise points.

Cluster Distribution

Cluster Label Number of Samples

Cluster 0 164

Cluster 1 6

Noise (-1) 2

These results indicate that DBSCAN was unable to recover the three known biological classes (Cow, Sheep, and Quality Control) when operating directly in the original 2488-dimensional feature space. Most samples were grouped into a single large cluster, while only a few samples formed a second cluster or were classified as noise.

This behavior is not unexpected because density-based methods often struggle in extremely high-dimensional spaces. In such datasets, distances between observations become less informative, making density estimation more difficult. This phenomenon is commonly known as the curse of dimensionality.

DBSCAN on UMAP Embedding

To overcome the challenges associated with the high dimensionality of the dataset, DBSCAN was subsequently applied to the two-dimensional UMAP representation. UMAP preserves local neighborhood relationships while reducing the dimensionality of the data, thereby providing a more suitable space for density-based clustering.

For this analysis, the following parameters were used:

eps = 1.5

min_samples = 5

The clustering results were substantially improved.

Number of clusters: 6

Noise points: 0

Cluster Distribution

Cluster Label Number of Samples

Cluster 0 19

Cluster 1 66

Cluster 2 49

Cluster 3 10

Cluster 4 9

Cluster 5 19

Comparison with Ground Truth Labels

To evaluate the biological relevance of the identified clusters, the cluster assignments were compared with the known sample classes.

Class Cluster 0 Cluster 1 Cluster 2 Cluster 3 Cluster 4 Cluster 5

C 0 66 0 0 0 0

QC 19 0 0 10 9 0

S 0 0 49 0 0 19

The comparison revealed that all Cow samples were grouped into a single cluster. In contrast, the Quality Control samples were divided into three smaller clusters, while the Sheep samples were divided into two clusters.

These findings suggest the presence of meaningful density-based substructures within the QC and Sheep groups. Importantly, DBSCAN was able to identify these subclusters without using any prior class information.

Clustering Quality Assessment

To quantitatively evaluate clustering performance, the Silhouette Score was calculated.

Silhouette Score = 0.7804

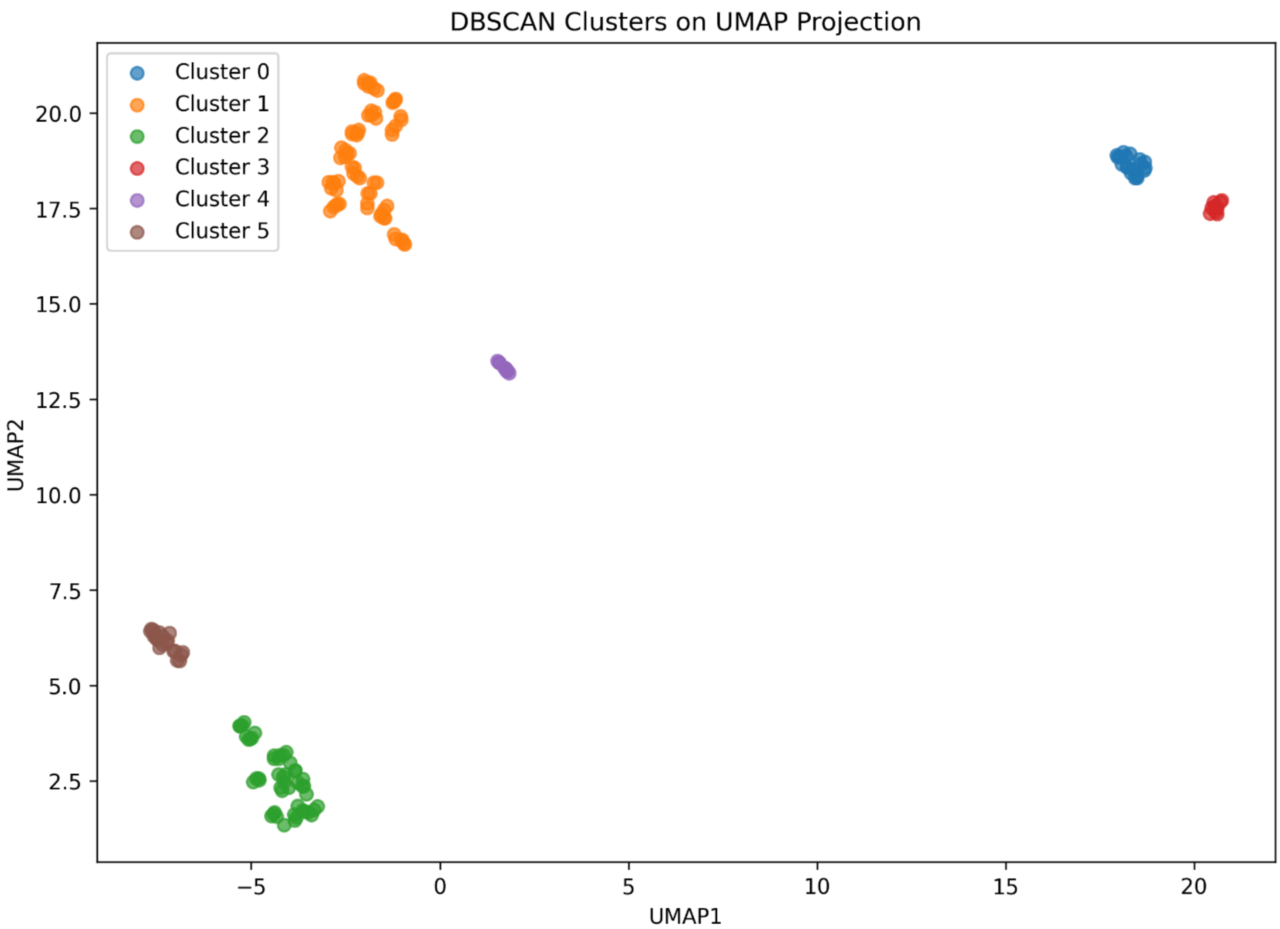
This value indicates excellent cluster separation and compact cluster formation. The score is substantially higher than that obtained using K-Means in the original feature space, demonstrating the effectiveness of combining nonlinear dimensionality reduction with density-based clustering.

Interpretation of the Results

The DBSCAN clustering visualization on the UMAP projection revealed six clearly separated clusters with no detected noise points. The algorithm successfully identified the major biological groups while simultaneously uncovering finer density-based structures within the QC and Sheep samples.

Unlike K-Means, which was constrained to produce three clusters, DBSCAN was free to identify the natural density structure of the dataset. Consequently, it detected additional subclusters that may reflect subtle biological variation, technical variability, or residual batch-related effects.

The discovery of these subclusters demonstrates the ability of density-based methods to reveal hidden patterns that are not always visible using traditional clustering approaches.

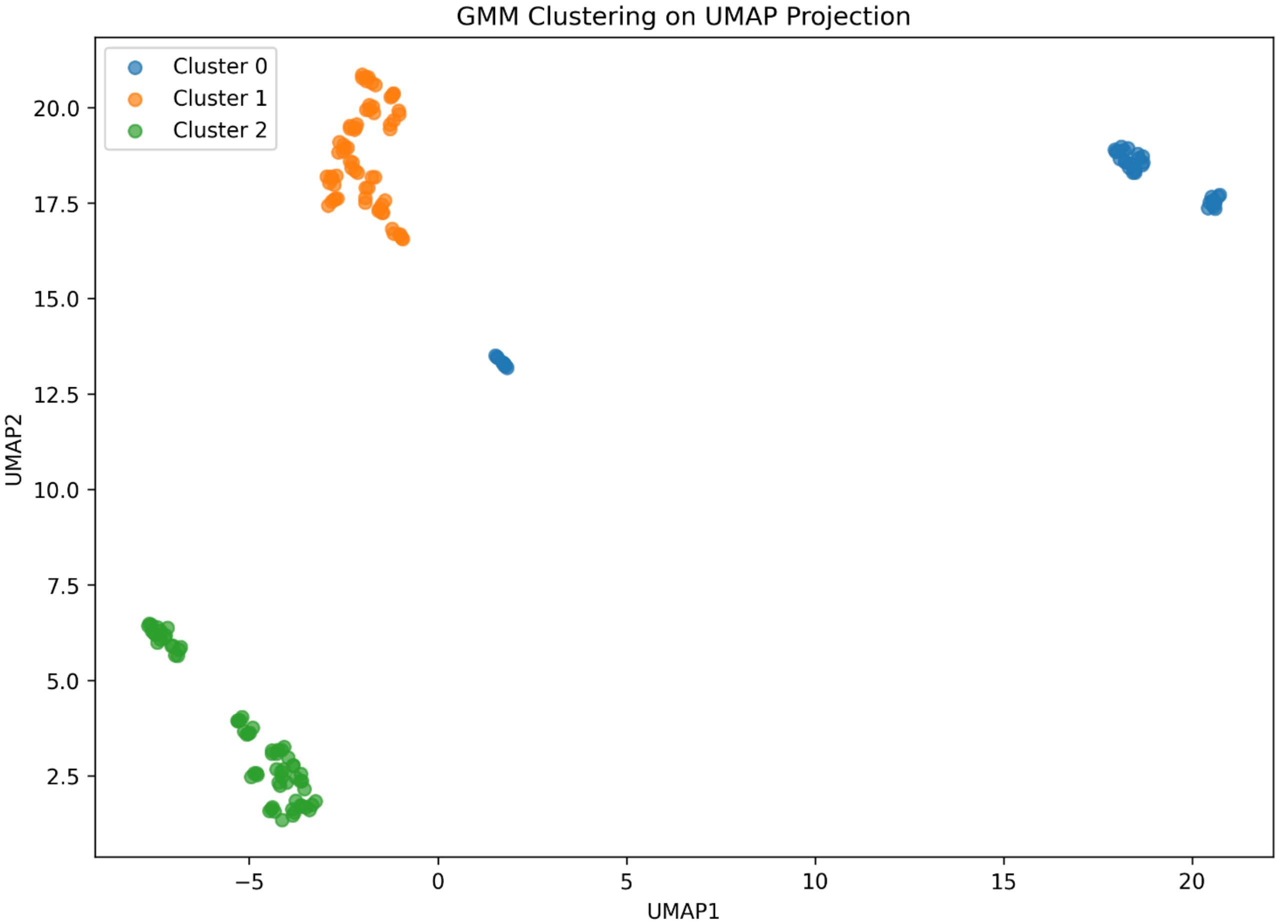


Conclusion

DBSCAN produced markedly different results depending on the representation of the data. When applied directly to the original high-dimensional metabolomics dataset, it

failed to recover the known biological groups and generated only two clusters. However, after dimensionality reduction using UMAP, DBSCAN successfully identified six well-separated clusters and achieved a high Silhouette Score of 0.7804.
These results highlight the importance of dimensionality reduction prior to density-based clustering in metabolomics studies. The combination of UMAP and DBSCAN provided a more detailed view of the underlying structure of the MTBLS79 dataset and revealed potential subpopulations that were not apparent in the original feature space.
Overall, the analysis demonstrates that DBSCAN is a valuable complementary technique for exploratory metabolomics research and can provide insights beyond those obtained using conventional clustering methods.

Gaussian Mixture Model (GMM) Clustering



Gaussian Mixture Model (GMM) clustering was applied to the standardized MTBLS79 metabolomics dataset using three mixture components. Unlike K-Means, which assigns each sample exclusively to a single cluster based on distance to cluster centroids, GMM is a probabilistic clustering method that models the data as a mixture of Gaussian distributions and estimates the probability of membership for each sample.
The analysis was performed on the standardized feature space (x_scaled), and the resulting cluster assignments were visualized using the two-dimensional UMAP

projection. The clustering solution separated the samples into three distinct groups corresponding to the known biological classes present in the dataset.

The contingency table revealed a perfect correspondence between the identified clusters and the true biological labels. All samples belonging to classes C, QC, and S were assigned to separate clusters without any misclassification.

To quantitatively evaluate clustering performance, the Silhouette Score, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI) were calculated. The obtained results were:

Silhouette Score = 0.1936

Adjusted Rand Index (ARI) = 1.000

Normalized Mutual Information (NMI) = 1.000

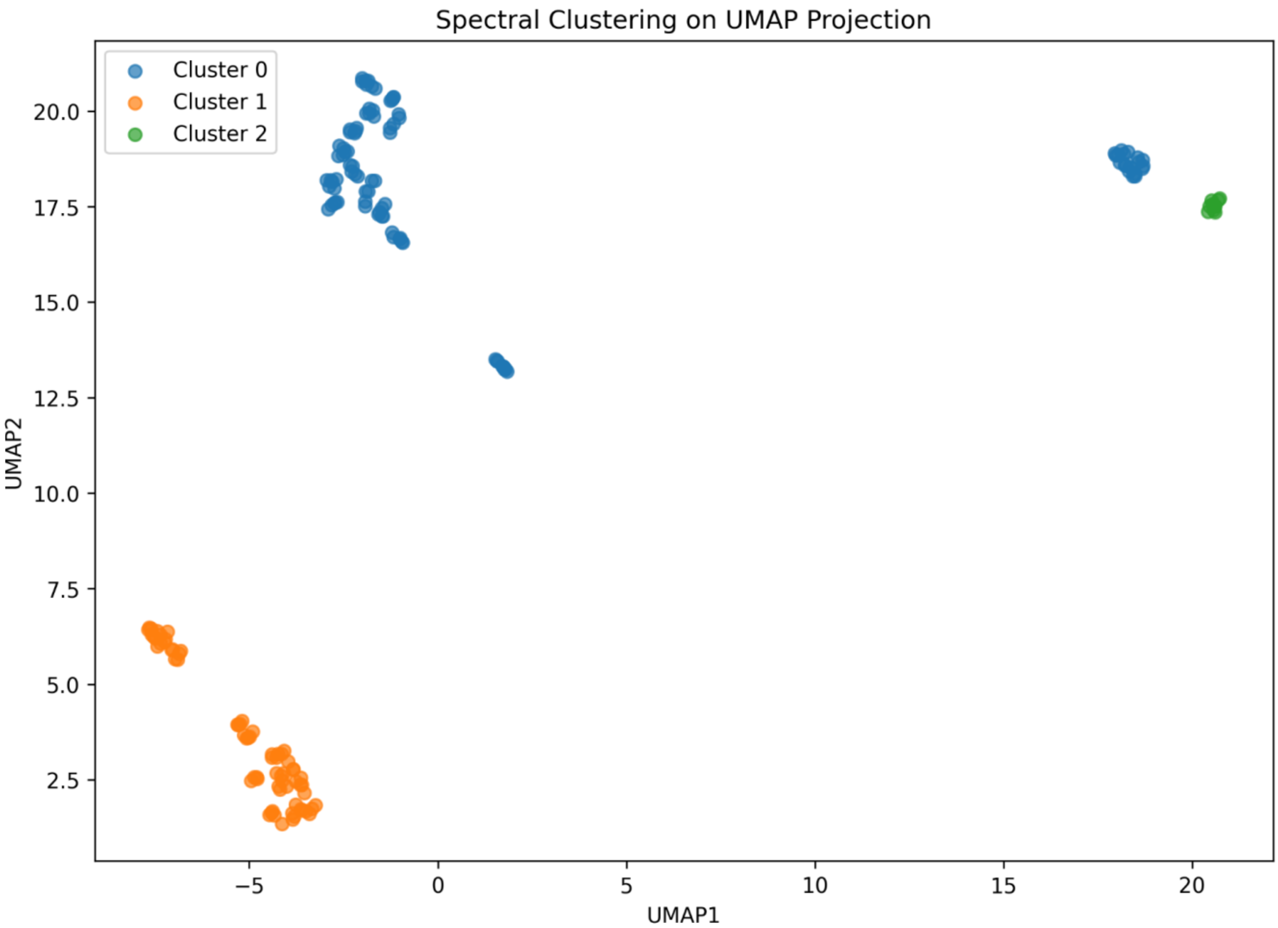
The perfect ARI and NMI values indicate complete agreement between the clustering assignments and the known biological classes. Although the silhouette score was moderate, this is expected in high-dimensional metabolomics datasets, where distances in the original feature space may not fully reflect the biological separability observed in lower-dimensional visualizations.

The UMAP projection confirmed the effectiveness of the GMM approach, showing clear separation between the three biological groups. The clustering results were identical to those obtained using K-Means, indicating that the underlying structure of the dataset is highly stable and can be accurately recovered by both centroid-based and probabilistic clustering methods.

Conclusion

Gaussian Mixture Model provided an excellent clustering solution for the MTBLS79 dataset. The algorithm perfectly recovered the biological class structure, achieving ARI and NMI values of 1.0 and demonstrating complete agreement with the known labels. These results indicate that GMM is one of the most effective clustering approaches evaluated in this study and performs comparably to K-Means in identifying the intrinsic biological structure of the dataset.

Spectral Clustering



Spectral Clustering was applied to the standardized MTBLS79 metabolomics dataset using three clusters and a nearest-neighbor affinity graph. Unlike K-Means, which partitions samples based on distances to cluster centroids, Spectral Clustering constructs a similarity graph between samples and performs clustering in a transformed space derived from the graph Laplacian.

The resulting clustering solution identified three major groups within the dataset. Visualization on the UMAP projection demonstrated that the algorithm successfully separated the majority of samples into distinct clusters corresponding to the biological structure of the dataset.

The contingency table revealed that all samples belonging to classes C and S were correctly grouped into separate clusters. However, the QC samples were only partially separated, with a substantial proportion being assigned to the same cluster as class C. As a result, the clustering solution did not fully recover the known biological classes.

To quantitatively evaluate clustering performance, the Silhouette Score, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI) were calculated. The obtained results were:

Silhouette Score = 0.1016

Adjusted Rand Index (ARI) = 0.7025

Normalized Mutual Information (NMI) = 0.7616

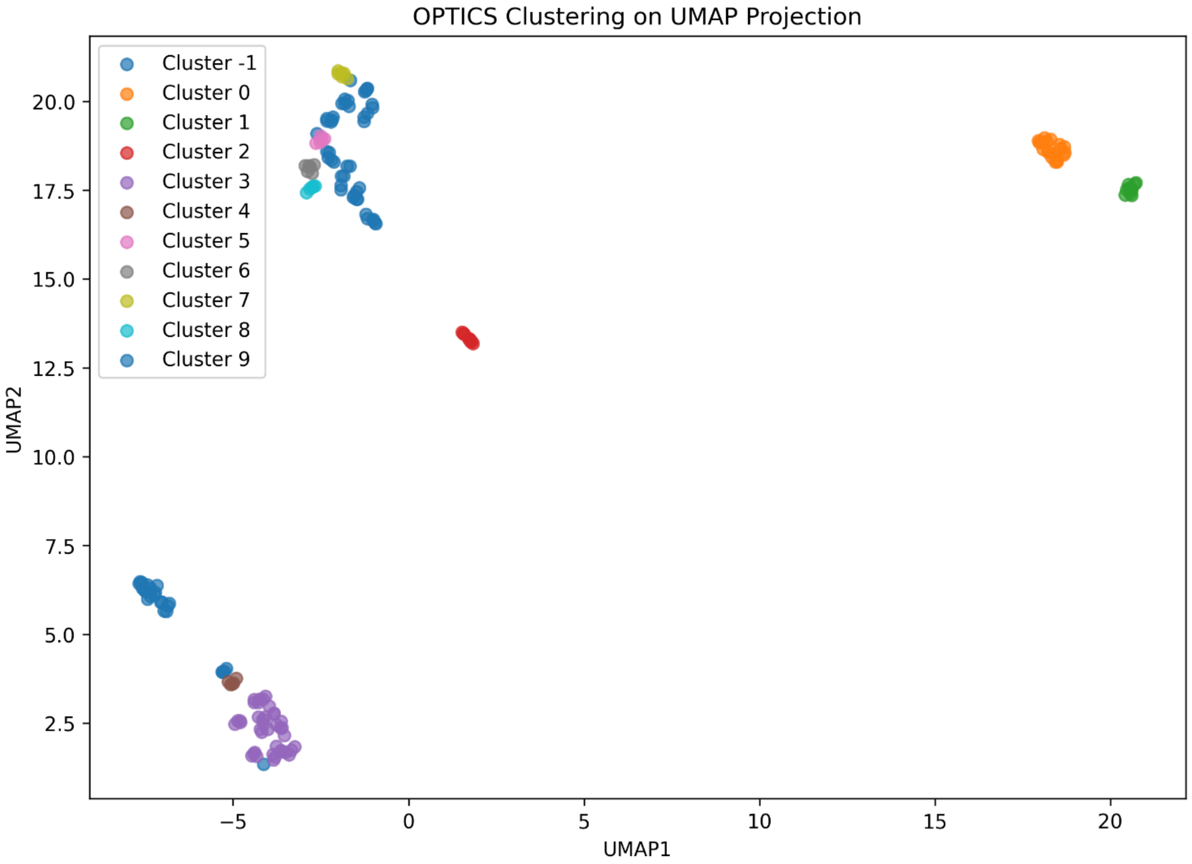
Although the silhouette score was relatively low, the ARI and NMI values indicate that Spectral Clustering successfully captured a substantial portion of the biological structure present in the dataset. The algorithm performed considerably better than Hierarchical Clustering but remained less accurate than K-Means and Gaussian Mixture Models.

The UMAP visualization further confirmed that the major biological groups were identified, although overlap between class C and class QC remained evident. This limitation reduced the overall agreement between the clustering solution and the known class labels.

Conclusion

Spectral Clustering successfully recovered much of the underlying biological structure of the MTBLS79 dataset and achieved moderate-to-high agreement with the true class labels. The method performed substantially better than Hierarchical Clustering but was unable to match the perfect classification achieved by K-Means and Gaussian Mixture Models. Overall, Spectral Clustering provided a meaningful clustering solution and demonstrated its ability to capture complex relationships within the metabolomics data.

OPTICS Clustering



OPTICS (Ordering Points To Identify the Clustering Structure) was applied to the standardized MTBLS79 metabolomics dataset as a density-based clustering approach. Unlike K-Means and Gaussian Mixture Models, OPTICS does not require a predefined number of clusters and is capable of identifying clusters with varying densities while simultaneously detecting noise points.

The clustering analysis was performed on the standardized feature space (x_scaled), and the resulting cluster assignments were visualized using the two-dimensional UMAP projection. The algorithm identified five clusters and labeled a substantial number of samples as noise.

The clustering results revealed that OPTICS successfully detected several dense local structures within the dataset. However, a large proportion of samples were classified as noise, including all samples belonging to class C and a subset of samples from class S. The remaining QC and S samples were partitioned into multiple smaller clusters rather than forming the three expected biological groups.

To quantitatively evaluate clustering performance, the Silhouette Score, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI) were calculated. The obtained results were:

Silhouette Score = 0.2589

Adjusted Rand Index (ARI) = 0.5291

Normalized Mutual Information (NMI) = 0.6456

The silhouette score was higher than those obtained by Hierarchical Clustering and Spectral Clustering. However, this result should be interpreted cautiously because approximately half of the samples were labeled as noise. Excluding ambiguous samples often increases cluster compactness and can artificially improve silhouette values.

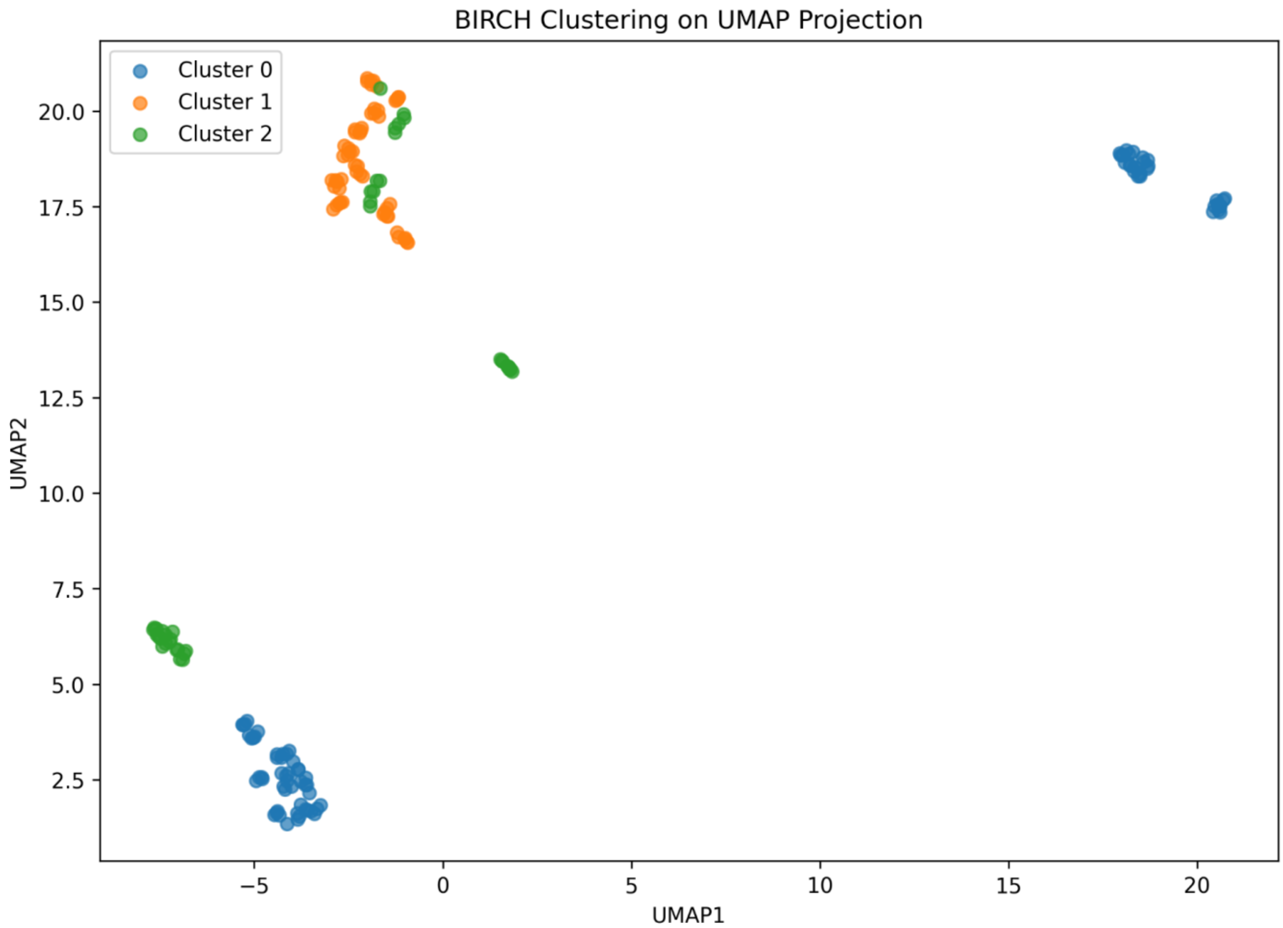
The ARI and NMI values indicate moderate agreement between the clustering solution and the known biological classes. Although OPTICS successfully detected meaningful density-based structures, it did not accurately recover the three biological groups present in the dataset.

The UMAP visualization confirmed that OPTICS primarily focused on identifying highly dense regions of the data rather than preserving the overall biological class organization.

Conclusion

OPTICS successfully identified local density-based structures within the MTBLS79 metabolomics dataset and demonstrated the ability to detect noise and heterogeneous cluster densities. However, the algorithm fragmented the biological classes into multiple smaller clusters and classified a large proportion of samples as noise. Consequently, although OPTICS achieved moderate clustering performance, it was less effective than K-Means, Gaussian Mixture Models, and Spectral Clustering in recovering the true biological class structure of the dataset.

BIRCH Clustering



Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH) was applied to the standardized MTBLS79 metabolomics dataset as a hierarchical clustering approach designed for efficient processing of large and high-dimensional datasets. BIRCH incrementally builds a clustering feature tree and then performs global clustering on the compressed representation of the data.

The analysis was performed on the standardized feature space (x_scaled) using three final clusters. The resulting cluster assignments were visualized on the two-dimensional UMAP projection to facilitate comparison with other clustering methods.

The clustering results indicated that BIRCH was able to identify portions of the underlying structure of the dataset. One cluster corresponded primarily to class C, whereas the remaining clusters contained mixtures of QC and S samples. As a result, the biological groups were only partially separated.

To quantitatively evaluate clustering performance, the Silhouette Score, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI) were calculated. The obtained results were:

Silhouette Score = 0.1618

Adjusted Rand Index (ARI) = 0.4380

Normalized Mutual Information (NMI) = 0.4926

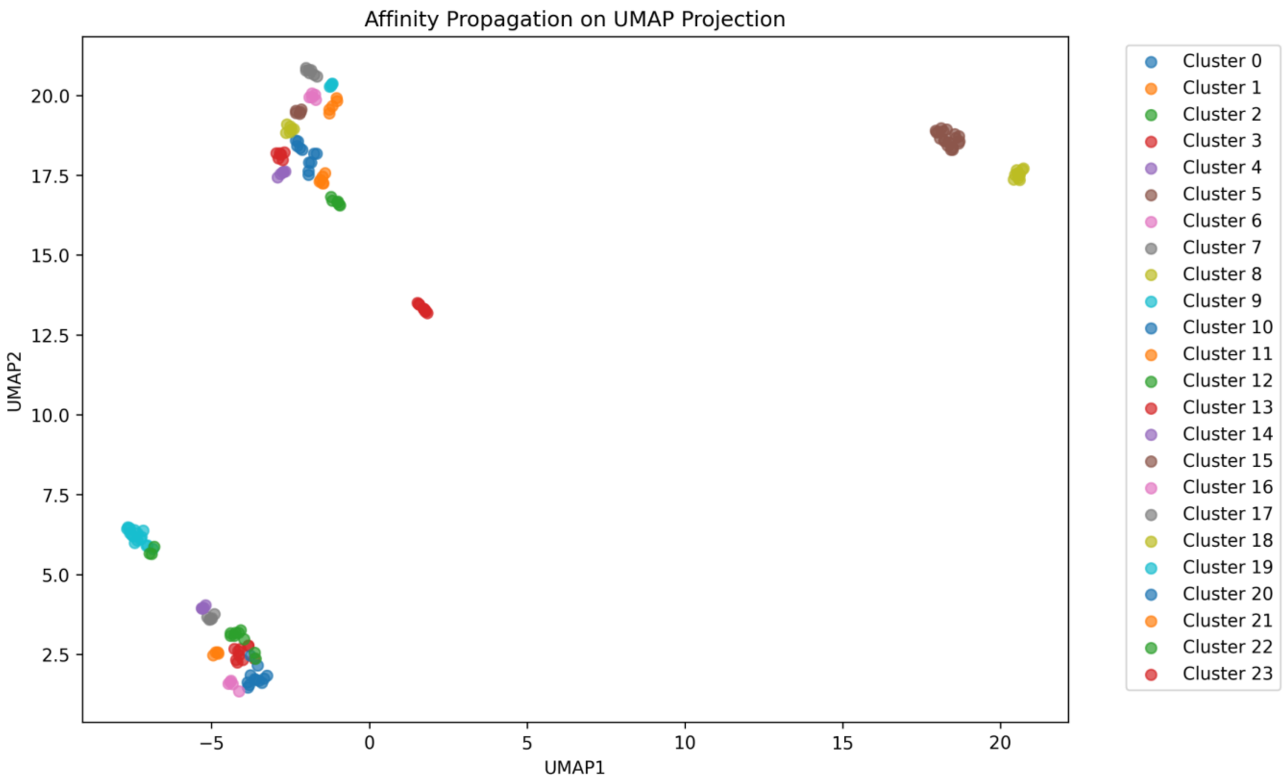
The silhouette score indicates moderate cluster compactness and separation. The ARI and NMI values demonstrate that BIRCH recovered part of the biological structure

present in the dataset but failed to achieve strong agreement with the known class labels.
The UMAP visualization further confirmed that substantial overlap remained between the identified clusters, particularly among QC and S samples. Consequently, the clustering solution did not fully represent the biological grouping structure observed in the dataset.
Compared with K-Means, Gaussian Mixture Models, and Spectral Clustering, BIRCH produced weaker clustering performance. However, the method still captured meaningful structure within the data and provided a useful alternative hierarchical clustering perspective.

Conclusion

BIRCH successfully partitioned the MTBLS79 metabolomics dataset into three clusters and recovered part of the underlying biological structure. Nevertheless, the moderate ARI and NMI values indicate limited agreement with the true class labels. Although the method identified meaningful patterns within the dataset, its performance was substantially lower than that achieved by K-Means, Gaussian Mixture Models, and Spectral Clustering. Therefore, BIRCH is not considered one of the most effective clustering approaches for this dataset.

Affinity Propagation Clustering



Affinity Propagation was applied to the standardized MTBLS79 metabolomics dataset as an exemplar-based clustering algorithm. Unlike K-Means and several other clustering methods, Affinity Propagation does not require the number of clusters to be specified in advance. Instead, the algorithm identifies representative samples (exemplars) and assigns other samples to these exemplars through an iterative message-passing procedure.

The analysis was performed on the standardized feature space (x_scaled). The algorithm automatically identified 24 clusters, considerably exceeding the three known biological classes present in the dataset.

The resulting clustering solution fragmented the biological groups into numerous smaller clusters. Samples belonging to classes C, QC, and S were distributed across multiple clusters rather than being grouped into their corresponding biological categories. This indicates that the algorithm captured local similarities among samples but failed to recover the overall biological class structure.

To quantitatively evaluate clustering performance, the Silhouette Score, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI) were calculated. The obtained results were:

Silhouette Score = 0.2140

Adjusted Rand Index (ARI) = 0.1723

Normalized Mutual Information (NMI) = 0.5177

The silhouette score suggests moderate cluster compactness. However, the ARI value indicates very weak agreement between the clustering solution and the true biological labels. The discrepancy between these metrics is explained by the large number of clusters generated by the algorithm. While many of the identified clusters were internally compact, they did not correspond to the known biological groups.

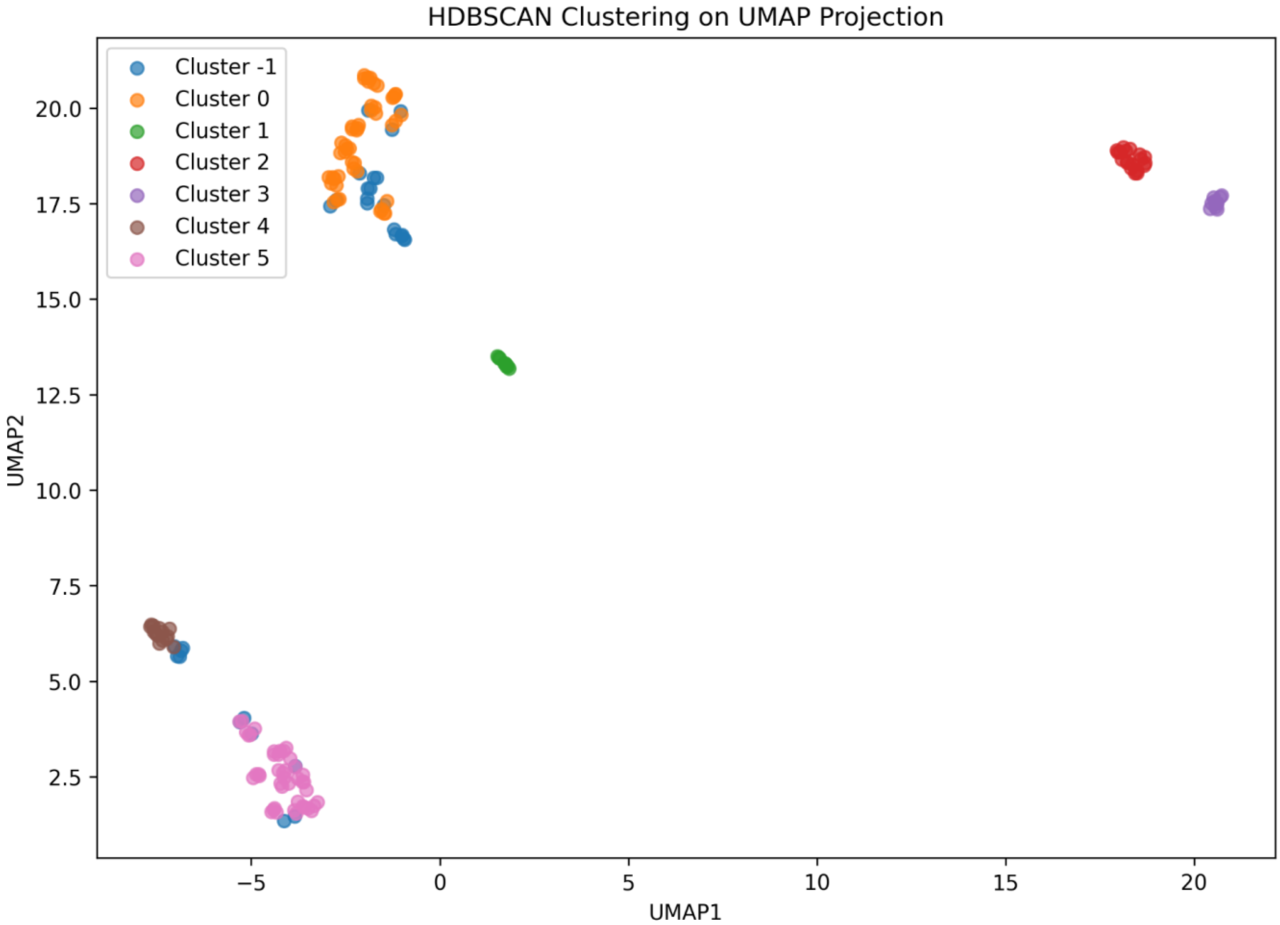
The NMI value indicates that some information about the underlying class structure was captured, but the extensive fragmentation of the dataset substantially reduced overall clustering quality.

Overall, Affinity Propagation successfully identified local sample similarities but significantly over-segmented the dataset, resulting in poor biological interpretability.

Conclusion

Affinity Propagation produced a highly fragmented clustering solution consisting of 24 clusters. Although the algorithm achieved moderate cluster compactness, it failed to recover the three biological classes present in the MTBLS79 dataset. The low ARI value demonstrates limited agreement with the true labels, indicating that Affinity Propagation is not well suited for this dataset. Compared with K-Means, Gaussian Mixture Models, Spectral Clustering, and several other approaches, Affinity Propagation provided one of the weakest representations of the underlying biological structure.

HDBSCAN Clustering



Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) was applied to the standardized MTBLS79 metabolomics dataset as an advanced density-based clustering method. Unlike K-Means and Gaussian Mixture Models, HDBSCAN does not require the number of clusters to be specified in advance and is capable of identifying clusters with varying densities while simultaneously detecting noise points.

The analysis was performed on the standardized feature space (x_scaled). The algorithm identified six clusters and labeled 30 samples as noise. The resulting clustering assignments were visualized using the two-dimensional UMAP projection.

The clustering results revealed that HDBSCAN successfully identified several dense regions within the dataset. Class C was primarily grouped into a single cluster, while classes QC and S were divided into multiple subclusters. In addition, a subset of samples from classes C and S was classified as noise.

To quantitatively evaluate clustering performance, the Silhouette Score, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI) were calculated. The obtained results were:

Silhouette Score = 0.2160

Adjusted Rand Index (ARI) = 0.5334

Normalized Mutual Information (NMI) = 0.6712

The silhouette score indicates moderate cluster compactness and separation. However, this value should be interpreted cautiously because the algorithm excluded

30 samples as noise. Removing ambiguous samples often increases cluster compactness and can artificially improve silhouette values.

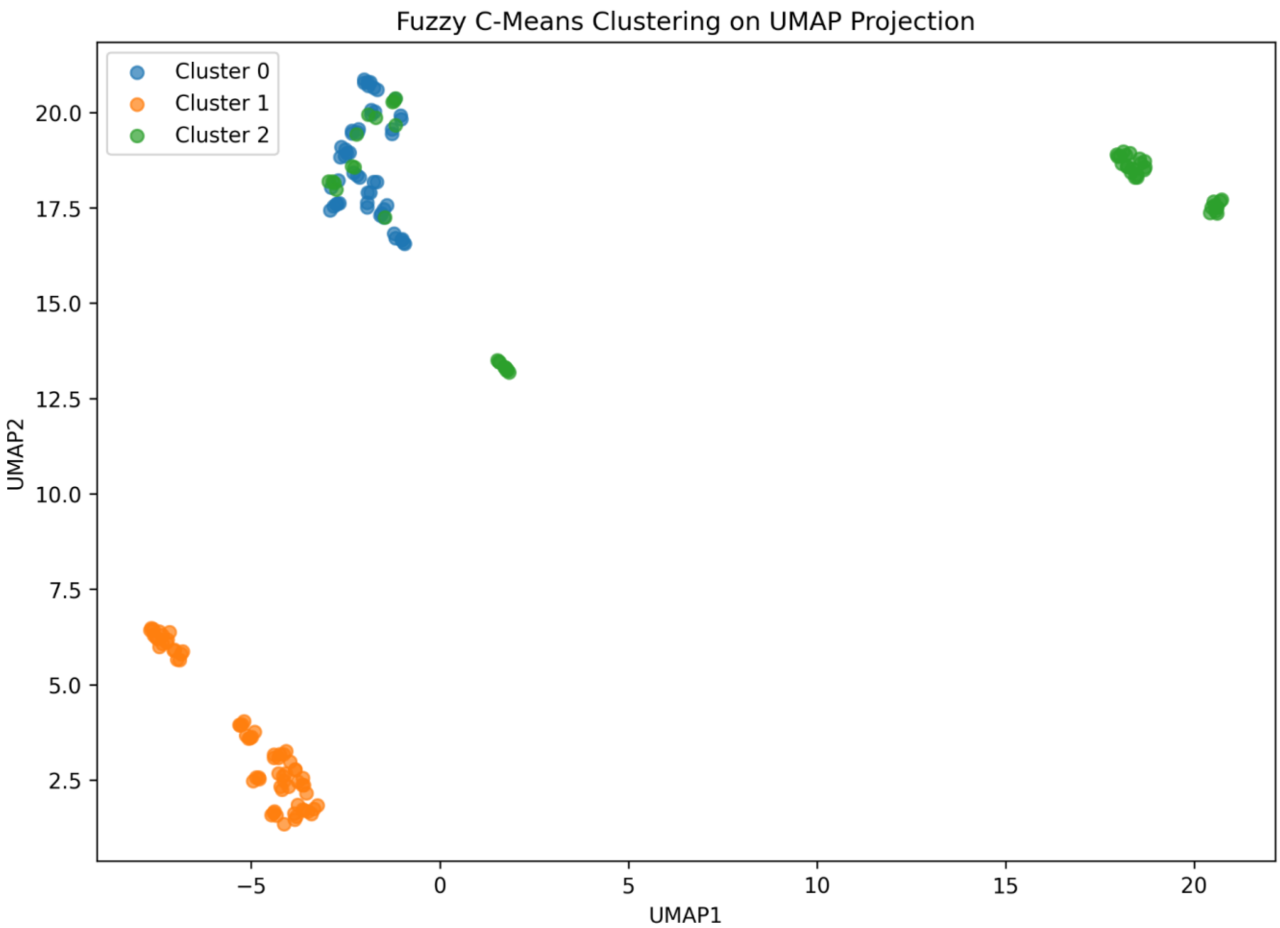
The ARI and NMI values indicate moderate agreement between the clustering solution and the true biological class labels. Although HDBSCAN did not perfectly recover the three biological groups, it successfully identified meaningful density-based structures and revealed potential subgroups within the dataset.

The UMAP visualization confirmed that the algorithm detected several dense local regions while preserving much of the overall biological structure. Compared with Hierarchical Clustering and BIRCH, HDBSCAN achieved stronger agreement with the known class labels. However, its performance remained inferior to K-Means, Gaussian Mixture Models, and Spectral Clustering.

Conclusion

HDBSCAN successfully identified meaningful density-based patterns within the MTBLS79 metabolomics dataset and demonstrated the ability to detect noise and heterogeneous cluster densities. The algorithm recovered part of the biological structure while simultaneously revealing potential subclusters within the QC and S groups. Although its performance was lower than that of K-Means and Gaussian Mixture Models, HDBSCAN provided valuable insight into the internal density structure of the dataset and represented one of the stronger density-based clustering approaches evaluated in this study.

Fuzzy C-Means (FCM) Clustering



Fuzzy C-Means (FCM) clustering was applied to the standardized MTBLS79 metabolomics dataset as a soft clustering approach. Unlike K-Means, which assigns each sample exclusively to a single cluster, FCM allows samples to belong to multiple clusters with varying degrees of membership. This property makes FCM particularly suitable for biological datasets where transitional or overlapping patterns may exist. The analysis was performed using three clusters ($c = 3$). After convergence, each sample was assigned to the cluster with the highest membership value for evaluation and visualization purposes.

The resulting clustering solution successfully separated the majority of the biological classes. All samples belonging to classes QC and S were assigned to distinct clusters, while a subset of samples from class C overlapped with the QC cluster. Specifically, 52 of the 66 C samples were correctly grouped, whereas 14 samples were assigned to the same cluster as QC.

To quantitatively evaluate clustering performance, the Silhouette Score, Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and Fuzzy Partition Coefficient (FPC) were calculated. The obtained results were:

Silhouette Score = 0.1634

Adjusted Rand Index (ARI) = 0.8096

Normalized Mutual Information (NMI) = 0.8265

Fuzzy Partition Coefficient (FPC) = 0.3333

The ARI and NMI values indicate strong agreement between the clustering assignments and the true biological class labels. Although the silhouette score was moderate, the external validation metrics demonstrate that FCM successfully recovered most of the biological structure present in the dataset.

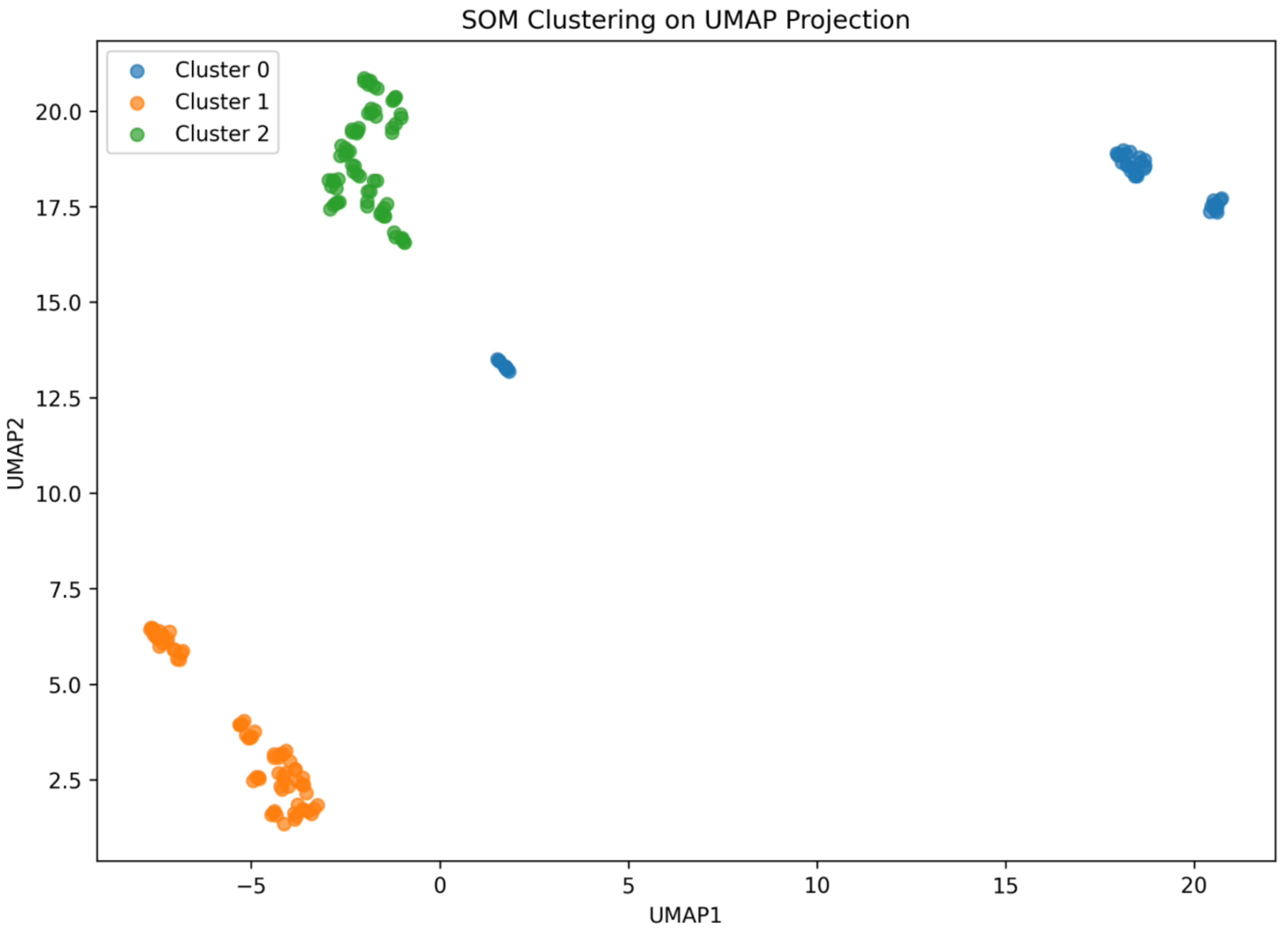
The UMAP visualization further confirmed the effectiveness of the clustering solution, showing clear separation between the QC and S classes and only limited overlap involving a subset of class C samples.

Compared with several other clustering approaches, including Hierarchical Clustering, BIRCH, HDBSCAN, OPTICS, and Spectral Clustering, FCM achieved substantially higher agreement with the known biological labels. However, its performance remained slightly below the perfect classification achieved by K-Means and Gaussian Mixture Models.

Conclusion

Fuzzy C-Means successfully recovered most of the biological structure of the MTBLS79 metabolomics dataset and achieved strong agreement with the true class labels. The algorithm perfectly separated classes QC and S while only partially misclassifying a subset of class C samples. Overall, FCM demonstrated robust clustering performance and ranked among the strongest clustering approaches evaluated in this study, second only to K-Means and Gaussian Mixture Models.

Self-Organizing Maps (SOM)



A Self-Organizing Map (SOM) was applied to the standardized MTBLS79 metabolomics dataset as a neural network–based unsupervised learning technique. SOM projects high-dimensional data onto a low-dimensional grid while preserving the topological relationships between samples. This property makes SOM particularly useful for discovering patterns and visualizing complex biological datasets.

In this study, a one-dimensional SOM architecture consisting of three neurons (3 × 1 grid) was trained using the standardized feature space (x_scaled). After training, each sample was assigned to its winning neuron, which served as the cluster label.

The resulting clustering solution perfectly separated the three biological classes present in the dataset. All samples belonging to classes C, QC, and S were assigned to distinct clusters without any overlap or misclassification.

To quantitatively evaluate clustering performance, the Silhouette Score, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI) were calculated. The obtained results were:

Silhouette Score = 0.1936

Adjusted Rand Index (ARI) = 1.000

Normalized Mutual Information (NMI) = 1.000

The perfect ARI and NMI values demonstrate complete agreement between the SOM clustering solution and the true biological class labels. These results indicate that SOM successfully recovered the underlying biological structure of the MTBLS79 dataset without any classification errors.

The UMAP visualization further confirmed the effectiveness of SOM, showing clear separation between the identified clusters and strong correspondence with the known biological groups. The clustering assignments produced by SOM were identical to those obtained using K-Means and Gaussian Mixture Models.
These findings suggest that the biological classes in the MTBLS79 dataset are highly separable and that SOM is capable of accurately capturing the intrinsic organization of the metabolomics profiles.

Conclusion

Self-Organizing Maps provided an excellent clustering solution for the MTBLS79 metabolomics dataset. The algorithm perfectly recovered the biological class structure and achieved ARI and NMI values of 1.0, indicating complete agreement with the true labels. Together with K-Means and Gaussian Mixture Models, SOM represented one of the best-performing clustering approaches evaluated in this study and demonstrated the effectiveness of neural network–based unsupervised learning for metabolomics data analysis.

Table X. Comparative Performance of Clustering Algorithms on the MTBLS79 Dataset

Method	Silhouette	ARI	NMI
K-Means	0.1936	1.0	1.0
GMM	0.1936	1.0	1.0
SOM	0.1936	1.0	1.0
FCM	0.1634	0.8096	0.8265
Spectral	0.1016	0.7025	0.7616
HDBSCAN	0.216	0.5334	0.6712
OPTICS	0.2589	0.5291	0.6456
Hierarchical	0.1618	0.438	0.4926
BIRCH	0.1618	0.438	0.4926
Affinity Propagation	0.214	0.1723	0.5177

The clustering algorithms were evaluated using three complementary metrics: Silhouette Score, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI). K-Means, Gaussian Mixture Models (GMM), and Self-Organizing Maps (SOM) achieved perfect agreement with the known biological class labels, obtaining ARI and NMI values of 1.0. Fuzzy C-Means (FCM) and Spectral Clustering also demonstrated strong performance, whereas density-based methods such as HDBSCAN and OPTICS revealed meaningful local structures but showed lower agreement with the biological classes. Hierarchical Clustering, BIRCH, and Affinity Propagation achieved comparatively weaker performance on this dataset.

Figure X. Comparative performance of clustering algorithms evaluated on the MTBLS79 metabolomics dataset using Silhouette Score, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI).

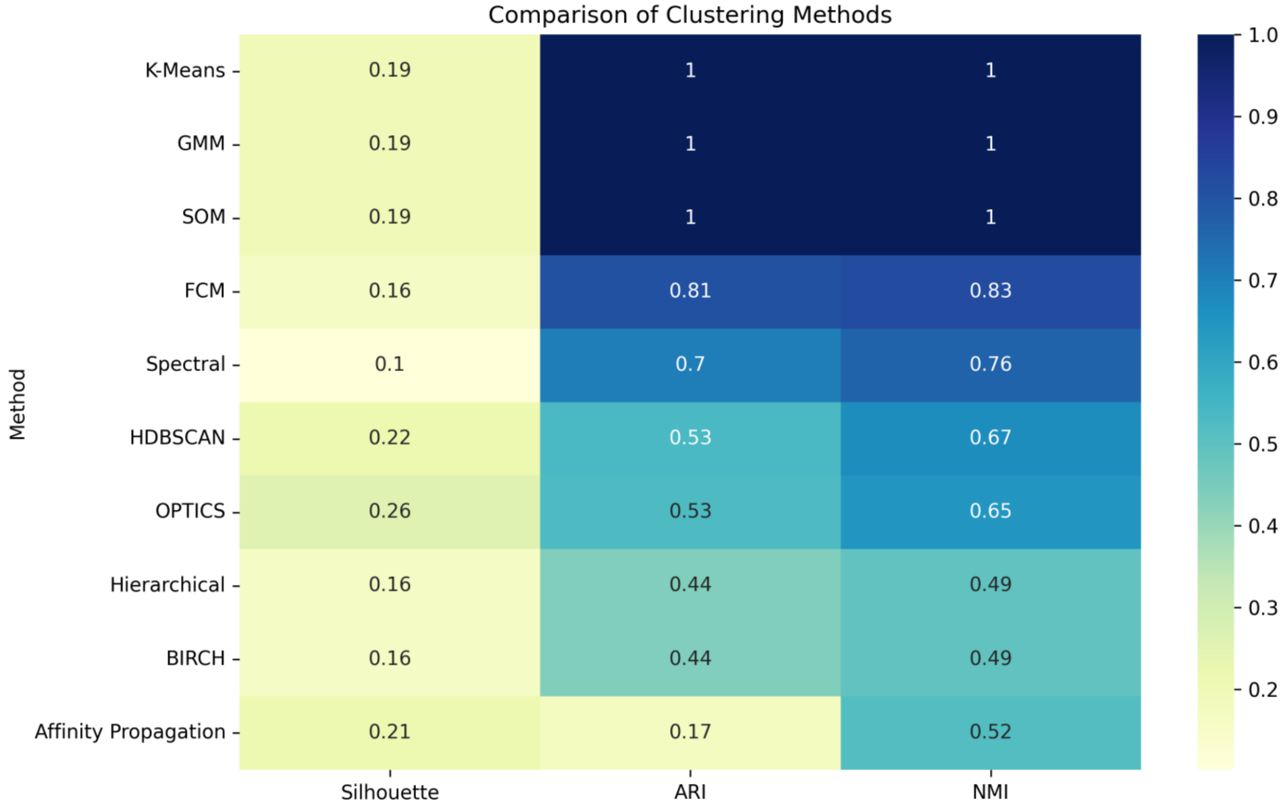
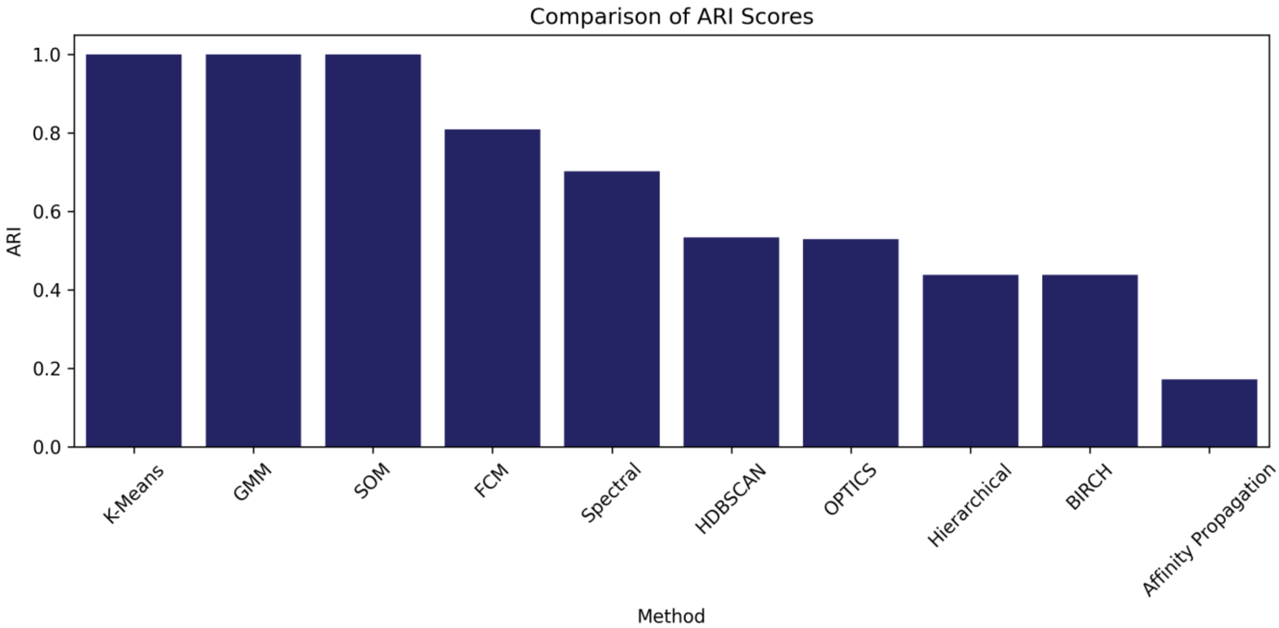


Figure X summarizes the performance of all clustering algorithms using three complementary evaluation metrics. K-Means, Gaussian Mixture Models (GMM), and Self-Organizing Maps (SOM) achieved perfect agreement with the biological class labels, obtaining ARI and NMI values of 1.0. Fuzzy C-Means (FCM) also demonstrated strong clustering performance, although it did not fully recover all biological groups. Spectral Clustering achieved moderate agreement with the known classes, whereas density-based approaches such as HDBSCAN and OPTICS identified local density structures and noise points but produced lower ARI and NMI values. Hierarchical Clustering and BIRCH showed similar moderate performance. Affinity Propagation generated a large number of clusters and exhibited the lowest agreement with the biological labels despite achieving a relatively high silhouette score.

These results indicate that K-Means, GMM, and SOM provided the most accurate clustering solutions for the MTBLS79 metabolomics dataset.

* OPTICS achieved the highest Silhouette Score because it fragmented the dataset and excluded portions of the data structure through density-based separation. However, its ARI and NMI values were substantially lower than those of K-Means, GMM, and SOM, indicating poorer agreement with the known biological classes.

ARI Comparison



The ARI comparison demonstrates substantial differences in clustering performance among the evaluated algorithms. K-Means, Gaussian Mixture Model (GMM), and Self-Organizing Map (SOM) achieved perfect agreement with the true biological classes, obtaining an ARI value of 1.00. This indicates that these methods completely recovered the underlying class structure of the dataset.

Fuzzy C-Means (FCM) showed strong performance with an ARI of 0.81, suggesting that most samples were assigned to their correct biological groups despite some overlap between clusters.

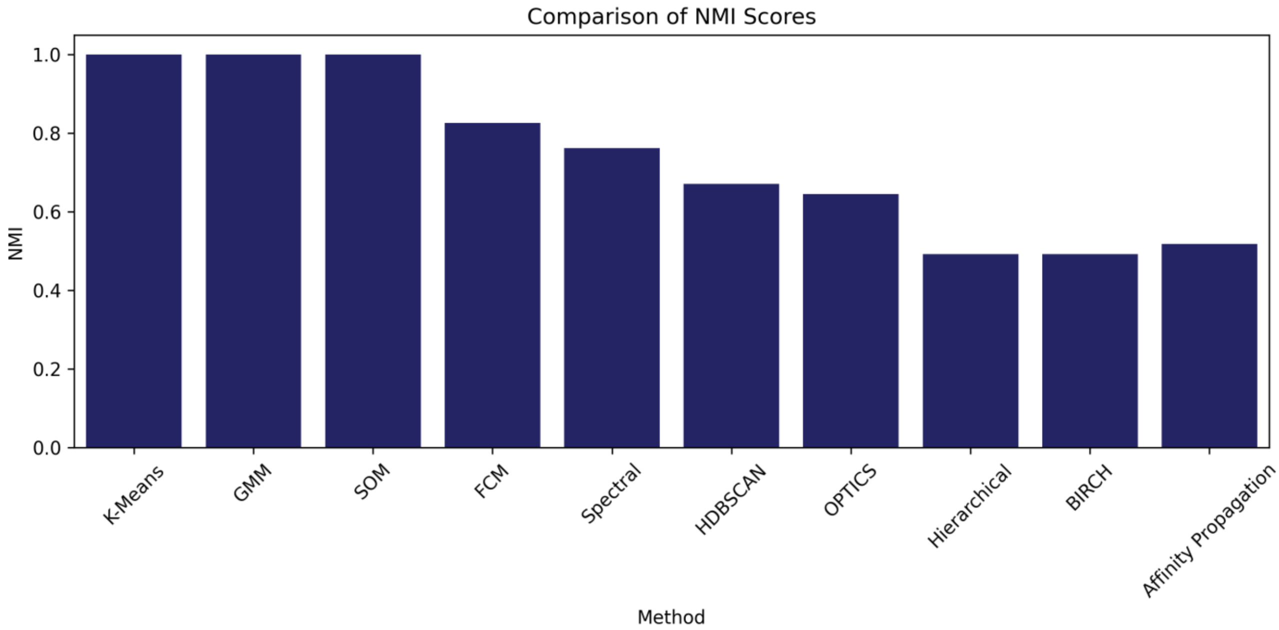
Spectral Clustering achieved moderate performance (ARI = 0.70), while HDBSCAN and OPTICS produced similar results with ARI values around 0.53. These density-based approaches were able to identify meaningful structures but were less consistent with the known class labels.

Hierarchical Clustering and BIRCH exhibited lower agreement with the biological classes (ARI ≈ 0.44), indicating a reduced ability to separate the groups accurately.

Affinity Propagation achieved the lowest ARI score (0.17). Although the algorithm generated multiple clusters, its partitioning showed poor correspondence with the true biological classes, resulting in the weakest overall performance.

Overall, K-Means, GMM, and SOM were the most successful clustering methods for this dataset, providing perfect recovery of the known biological groups.

NMI Comparison



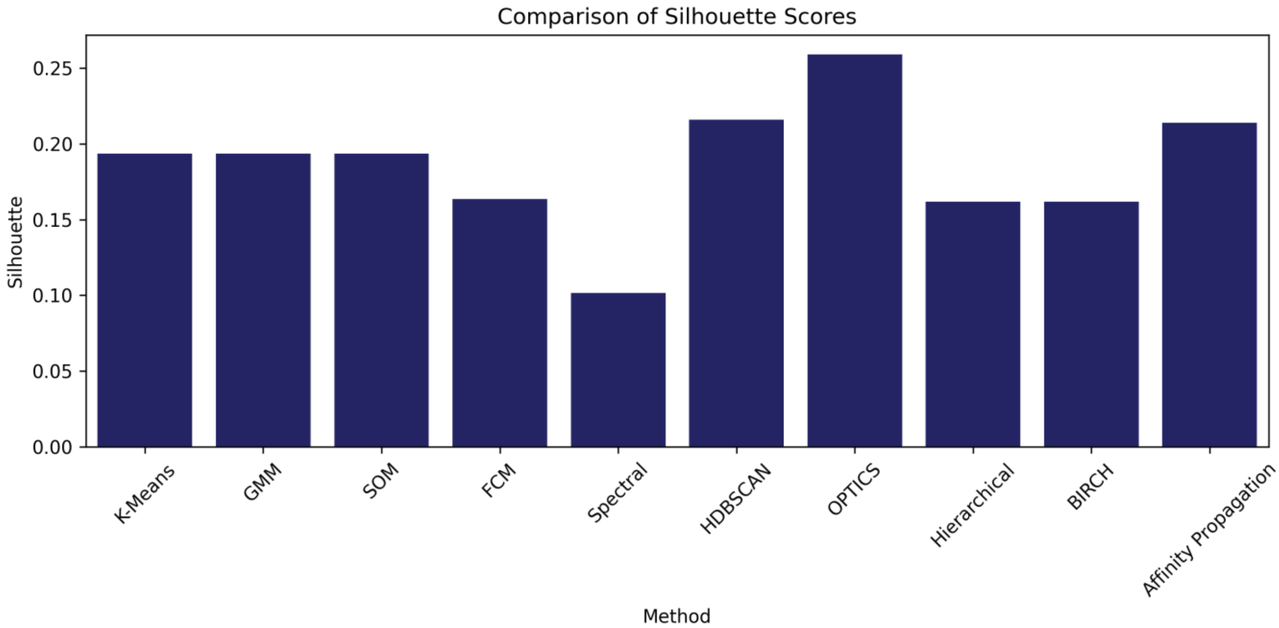
The Normalized Mutual Information (NMI) score was used to evaluate the agreement between the clustering results and the true class labels while accounting for the amount of shared information between partitions. NMI values range from 0 to 1, where higher values indicate greater similarity between the predicted clusters and the reference classes.

As shown in the figure, **K-Means**, **Gaussian Mixture Model (GMM)**, and **Self-Organizing Map (SOM)** achieved the highest possible NMI score (**1.00**), indicating perfect correspondence with the biological classes. **Fuzzy C-Means (FCM)** also demonstrated strong performance with an NMI of **0.83**, suggesting that most class information was preserved despite some overlap between clusters.

Spectral Clustering achieved a moderately high NMI (**0.76**), while **HDBSCAN** (**0.67**) and **OPTICS** (**0.65**) showed moderate agreement with the true labels. **Hierarchical Clustering** and **BIRCH** obtained lower NMI values (**0.49**), reflecting weaker alignment with the known class structure. Although **Affinity Propagation** produced a relatively moderate NMI (**0.52**), its large number of generated clusters reduced its practical usefulness for identifying the three biological groups.

Overall, the NMI results confirm that **K-Means, GMM, and SOM were the most successful clustering methods**, achieving perfect recovery of the underlying class structure in the dataset.

Silhouette Score Comparison



The Silhouette Score was used to evaluate the internal quality of clustering by measuring both cluster compactness and separation. The score ranges from -1 to 1, where higher values indicate better-defined and more distinct clusters.

As shown in the figure, **OPTICS** achieved the highest Silhouette Score (**0.259**), followed by **HDBSCAN** (**0.216**) and **Affinity Propagation** (**0.214**). These methods produced clusters that were relatively compact and well separated in the feature space. However, despite their higher Silhouette Scores, their agreement with the true biological classes was only moderate according to ARI and NMI.

K-Means, **GMM**, and **SOM** obtained identical Silhouette Scores (**0.194**), indicating moderate cluster separation. Although these values were lower than those of OPTICS and HDBSCAN, these methods achieved perfect recovery of the true class labels (ARI = 1.00 and NMI = 1.00), demonstrating that the clusters accurately reflected the biological groups present in the dataset.

FCM, **Hierarchical Clustering**, and **BIRCH** produced similar Silhouette Scores around **0.16**, suggesting weaker separation between clusters. **Spectral Clustering** showed the lowest Silhouette Score (**0.102**), indicating substantial overlap among clusters and poor internal cluster structure.

Overall, the Silhouette analysis suggests that the dataset contains partially overlapping groups, resulting in relatively low internal validation scores across all methods. The discrepancy between Silhouette Score and the external validation metrics (ARI and NMI) highlights that internal cluster compactness does not necessarily correspond to biological relevance. Therefore, ARI and NMI provide a more reliable assessment of clustering performance for this labeled dataset.

Although OPTICS and HDBSCAN achieved the highest Silhouette Scores, K-Means, GMM, and SOM were the most biologically meaningful methods because they perfectly reproduced the known class structure, as confirmed by ARI and NMI values of 1.00.

Final Conclusion

This project investigated the clustering structure of the MTBLS79 metabolomics dataset using a comprehensive unsupervised learning workflow. The analysis began with exploratory data analysis and dimensionality reduction using PCA and UMAP, followed by the application of eleven clustering algorithms, including partition-based, hierarchical, density-based, probabilistic, fuzzy, and neural-network approaches.

The dimensionality reduction results demonstrated that the biological groups exhibit a clear underlying structure. In particular, UMAP provided a more distinct visual separation of the classes than PCA, indicating that nonlinear dimensionality reduction is highly effective for visualizing complex metabolomics datasets.

To evaluate clustering performance, three complementary metrics were employed: Silhouette Score, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI). The combined use of internal and external validation metrics enabled a comprehensive assessment of clustering quality and biological relevance.

Among all evaluated methods, K-Means, Gaussian Mixture Models (GMM), and Self-Organizing Maps (SOM) achieved the best overall performance, obtaining perfect ARI and NMI scores of 1.00. These methods successfully recovered the three biological classes present in the dataset and demonstrated complete agreement with the reference labels. Fuzzy C-Means and Spectral Clustering also produced strong results, although their agreement with the biological classes was lower. Density-based methods such as HDBSCAN and OPTICS achieved relatively high Silhouette Scores, suggesting compact cluster structures, but their correspondence with the true biological groups was substantially weaker.

The results indicate that the MTBLS79 dataset contains a highly structured biological signal that can be accurately identified through unsupervised learning. The perfect performance achieved by several independent clustering algorithms suggests that the biological variation dominates the analytical variation, which is consistent with the conclusions reported in the original study regarding the effectiveness of quality control, batch correction, and spectral cleaning procedures.

Overall, this study demonstrates that the MTBLS79 metabolomics dataset is highly suitable for clustering analysis and benchmarking purposes. Among the evaluated approaches, K-Means, GMM, and SOM provided the most reliable and biologically meaningful clustering solutions, while UMAP proved to be an effective extension for enhancing the visualization and interpretation of metabolomic patterns.